LEYUAN ANIMAL HUSBANDRY WEIXIAN CO., LTD. NO. 4 PASTURE WASTES RECYCLING PROJECT

Document Prepared by China Classification Society Certification Co. Ltd.

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Project Title	Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture Wastes Recycling Project
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Summary:

- A description of the project

China Classification Society Certification Co. Ltd. (CCSC), commissioned by Climate Bridge (Shanghai) Ltd. has performed the validation of Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture Wastes Recycling Project (hereafter referred to as “the project”) and the verification of the project covering the monitoring period from 21-Mar-2021 to 30-Sep-2022, on the basis of requirements of Verified Carbon Standard (VCS) Version 4.4.

The project is to utilize animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters and the recovered biogas for electricity generation. The project involves the construction and operation of the anaerobic animal manure management system, the biogas pre-treatment system,the organic fertilizer production plant and the biogas power generation system at Xiwang Village, Houguan Town, Weixian District, Xingtai City, Hebei Province, P.R China. According to the feasibility study report, the intended installed

capacity of the electricity generator is 3.83MW (1.165MW * 2 + 1.5MW *1). The project is divided into two phases. In phase Ⅰ, the installed capacity is 2.33MW (1.165MW * 2) and in phase Ⅱ, the installed capacity will be 1.5MW. The construction of phase Ⅱ will depend on the circumstances in the near future. The project is expected to achieve an average annual emission reduction of 57,220 tCO_2e and a total emission reduction of 400,540 tCO_2e during the first 7-year crediting period, and supply 27,000 MWh of electricity per year to the North China Power Grid (NCPG). During this monitoring period from 21-Mar-2021 to 30-Sep-2022, the actual operating capacity of the project is 2.33MW (2 sets of 1.165MW CMM generators have been operated to generate electricity), the achieved emission reductions are 47,385 tCO_2e.

The project displaces the equivalent power generated by the existing power plants and likely capacity additions within the given grid and avoids GHG emissions of methane through recovery and destruction of biogas.

- A description of the validation and verification

The approved CDM methodology AMS-III.D "Methodology Methane recovery in animal manure management systems" (Version 21.0) and AMS-I.D"Grid connected renewable electricity generation" (Version 18.0) are applied to quantify the GHG removals achieved in this project. The calculation of the GHG emission reductions is carried out in a transparent and conservative manner.

This project is being developed in conjunction with the validation and 1st periodic of verification.

- The purpose and scope of validation and verification

The validation objective is an independent assessment by a Third Party of a proposed project activity against all defined criteria set for the registration under the VCS. In order to confirm that the project activity, as documented, is sound reasonable and meets the identified criteria, the validation involves the assessment of: project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCUs. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to VCS registry.

The objective of the verification is to have an independent review ex post determination by a VVB (Validation and Verification Body) of the monitored GHG emission reductions that have occurred as a result of the implementation of the project activity during a defined monitoring period.

- The method and criteria used for validation and verification

Validation and Verification is conducted using China Classification Society Certification Co. Ltd. (CCSC) procedures in line with the requirements specified in the latest version of the VCS Validation and Verification Manual and applying auditing techniques. The validation/verification team assessed the project activity's compliance against the VCS Standard Version 4.4, the selected CDM methodology and the joint project description and monitoring report. The project is eligible under Sectoral Scope 1 and Sectoral scope 13. The validation/verification criteria followed the guidance documents provided by VCS included the following:

VCS Standard Version 4.4, VCS Program Guide Version 4.3 and the applied CDM methodology AMS-III.D "Methodology Methane recovery in animal manure management systems" (Version 21.0) and AMS-I.D"Grid connected renewable electricity generation" (Version 18.0)

- The number of findings raised during validation and verification

A risk-based approach has been followed to perform this validation and verification activity. CCSC appointed a qualified team in accordance with internal procedures to perform the validation and verification. During the validation and verification, 0 Clarification Request (CL), 2 Corrective Action Requests (CARs) and no forward action request (FAR) were raised.

- Any uncertainties associated with the validation and verification

There are no restrictions of uncertainty for both validation and verification.

- Summary of the validation and verification conclusions

Climate Bridge (Shanghai) Ltd. has commissioned the China Classification Society Certification Co. Ltd. (CCSC) to carry out the Verified Carbon Standard (VCS) validation joint with 1st periodical verification of the project, "Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture Wastes Recycling Project " (VCS ID 3984), with regard to the relevant requirements of VCS standard Version 4.4. CCSC confirms all validation and verification activities including objectives, scope and criteria, level of assurance, project description, monitoring and monitoring report adhere to VCS Standard Version 4.4 and all associated updated as documented in this report, are complete.

CCSC concludes that the project activity "Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture Wastes Recycling Project" in China, as described in the final version of Joint-Project-Description-Monitoring-Report, meets all relevant requirements for VCS validation and verification activity and correctly applied the methodology AMS-III.D (Version 21.0) and AMS-I.D (Version 18.0). CCSC is able to certify that the GHG emission reductions from the project during the monitoring period from 21-March-2021 to 30-September-2022 were generated as 47,385 tCO_2e which can be claimed as VCUs.

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1 INTRODUCTION

1.1 Objective

Climate Bridge (Shanghai) Ltd. has commissioned the China Classification Society Certification Co. Ltd. (CCSC) to carry out the Verified Carbon Standard (VCS) validation joint with 1st periodical verification of the project for the monitoring period from 21-March-2021 to 30-September-2022 (both days included).

The objective of validation is an independent assessment by a Third Party of a proposed project activity against all defined criteria set for the registration under the VCS. In order to confirm that the project activity, as documented, is rational and meets the identified criteria, the validation involves the assessment of: project conformance to VCS standards/programs, project conformance to the applied methodology, including the procedure for the demonstration of additionality specified in the methodology; and likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented. Validation is a requirement and is seen as necessary to provide assurance to stakeholders of the quality of project and its intended generation of VCUs. Validation is part of the VCS project cycle and will finally result in a conclusion by the executing VVB whether a project activity is valid to be submitted for registration to VCS registry.

The objective of verification is to have an independent review ex-post determination by a VVB of the monitored GHG emission reductions that have occurred as a result of the project activity implemented during a defined monitoring period. The evaluation is done against the requirements of the VCS Standard Version 4.4 based on the monitoring report. In order to confirm that the GHG emission reductions are reasonably quantified and meet the identified criteria, the verification involves the assessment of monitoring plan conformance to VCS rules and applied methodology. Verification is a requirement and is seen as necessary to provide assurance on the generation of VCUs.

1.2 Scope and Criteria

The validation was performed on the basis of VCS criteria. The information in these documents is reviewed against CDM Validation and Verification Standard, VCS standard version 4.4 and associated interpretations. CCSC has, based on the instructions in the VVS employed a risk-based and step-wise approach when conducting the validation, focusing on the identification of significant risks for project implementation and the generation of VCUs. The validation is not meant to provide any consulting towards the PPs. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The verification scope encompasses an independent and objective review and ex-post determination of the monitored reductions in GHG emissions by the CCSC. The verification scope covers the relevant documents (e.g. the Joint PD-MR /1//3/, the monitoring plan, the emission reduction(ER) calculation spreadsheet /2//4/, supporting documents available to the assessment team, and information collected through performing interviews and during the on-site inspection, VERRA's requirements publicly available, relevant rules, including the host country legislation, etc.) to be independently reviewed, the project geographical locations to be visited on-site, the related project stakeholders to be interviewed with, and processes that are necessary to acquire objective evidence for the evaluation of the project compliance to the VCS requirements and associated interpretations. The verification activities are conducted according to the VERRA's requirements. In doing so, the principles of accuracy and completeness, relevance, reliability and credibility are followed. The verification is not meant to provide any consulting service towards the PPs. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the project.

1.3 Reasonableness of Assumptions and Level of Assurance

CCSC has undertaken a reasonable assurance engagement in accordance with VCS Standard (version 4.4) /5/. The validation and verification conclusions are based on the Joint PD -MR /3/ and supporting evidence made available to the VVB and information collected through performing interviews and during the on-site inspection.

1.4 Summary Description of the Project

The project is owned and implemented by Hebei Zhonghai Huaneng Energy Co., Ltd. The project is to utilize animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters and the recovered biogas is for electricity generation. The project involves the construction and operation of the anaerobic animal manure management system, the biogas pre-treatment system, the organic fertilizer production plant and the biogas power generation system at Xiwang Village, Houguan Town, Weixian District, Xingtai City, Hebei Province, P.R China. According to the feasibility study report, the intended installed capacity of the electricity generator is 3.83MW (1.165MW * 2 + 1.5MW *1). The project is divided into two phases. In phase I, the installed capacity is 2.33MW (1.165MW*2) and in phase Ⅱ, the installed capacity will be 1.5MW. The construction of phase Ⅱ will depend on the circumstances in the near future.

During this monitoring period from 21-Mar-2021 to 30-Sep-2022, the actual operating capacity of the project is 2.33MW (2 sets of 1.165MW CMM generators have been operated to generate electricity). The centre geographical coordinates for the project site are east longitude 115°26′20.24″ and north latitude 37°07′58.57″.

The project started construction on 07-December-2019 and started operation on 21-March-2021. The project is expected to supply 27,000 MWh of electricity per year to the North China Power Grid (NCPG), displaces the equivalent power generated by the existing power plants and likely capacity additions within the given grid and avoids GHG emissions of methane through recovery and destruction of biogas.

The project includes the anaerobic animal manure management system, the biogas pre-treatment system, the organic fertilizer production plant and the biogas power generation system. An emergency flare system is installed at the project site, this emergency flare system is not used under normal operation. In case this emergency flare system operated, the emissions reductions during this period will be excluded for conservativeness. And VCUs will only be claimed from the methane destroyed by power generation system and the power displacement. The project emissions from flaring system are excluded for simplification.

The total amount of emission reductions are 400,540 tCO_2e during the first 7-year crediting period from 21-March-2021 to 20-March-2028, and the average annual emission reductions are 57,220 tCO_2e.

During the 1st monitoring period from 21-Mar-2021 to 30-Sep-2022, the project has been operated normally. The total GHG emission reductions generated by the project are verified to be 47,385 tCO_2e.

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2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

A project specific validation and verification plan was developed to guide the validation and verification auditing process to ensure efficiency and effectiveness.

The purpose of the validation and verification is to present a risk assessment for determining the nature and extent of validation and verification procedures necessary to ensure the risk of auditing error is reduced to a reasonable level. According to the ISO14064-3, the criteria are the policy, procedure or requirement used as reference against which evidence is compared. Therefore, validation of the project description and verification of the monitoring plan and the reported project results were measured for compliance against the following criteria:

1) VCS Standard, v4.4/5/

2) VCS Program Guide, v4.3/12/

3) VCS Program Definitions, v4.3/36/

4) VCS Methodology Requirements, v4.3/14/

5) VCS-Joint-Project-Description-Monitoring-Report-Template-v4.2/37/

The validation and verification process derived from all items in the validation and verification criteria stated above. Field inspection and techniques based on the project parameters, scope and best professional judgement of the validation and verification team in order to meet a reasonable level of assurance. The validation and verification consisted of the following three phase:

1) Document review

2) On-site assessment

3) The resolution of outstanding issues and the issuance of the final joint validation and

verification report and certification.

No sampling was utilized during the on-site inspection as well as validation and verification for project activity.

The following sections outline each step in more detail.

2.2 Document Review

A desk review of the Joint PD-MR/1//3/ and supporting documents was conducted by the assessment team. The aim of the desk review of the documentation was to verify correctness, credibility, interpretation and completeness of presented data and information, and to cross check between information provided in the Joint PD-MR/1//3/ and information from sources other than that used, if available. Review of the appropriateness of formulae and correctness of calculations was also carried out during this stage based on the approved methodology being applied. And particular attention was given to the quality of the metering equipment including calibration requirements, the monitored data and its evidence, and the emission reduction calculation. The evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions was also conducted.

Appendix B of this report contains a complete list of all documents and proofs reviewed by the assessment team.

2.3 Interviews

On 04-March-2023, CCSC performed an on-site inspection at the physical site of the project.

During the on-site inspection, the team has interviewed with key personnel from the project owner, local environmental protection bureau officer, and the consultancy. The assessment team also visited the local village to interview the local stakeholder about the project status randomly.

The assessment content and topics/the persons interviewed at the on-site inspection were provided in the below table.

Date: 04-March-2023	
Interview topics	**Interviewed Organizations and persons**
--The status of VCS project. --The evidences of construction and operation of key equipment, --Main equipment and technical parameters --Project boundary. --Monitoring plan, monitoring device installation and calibration --Data processing activities, --Monitoring data collection. --Quality Management; organizational structure, responsibilities and competencies; Internal QA/QC Management procedures and document control.	**Hebei Zhonghai Huaneng Energy Co., Ltd. (Project Owner)** Li Jihui, General manager assistant Li Mei, Deputy Office Director Ding Zhe, Operating staff

Date: 04-March-2023	
Interview topics	**Interviewed Organizations and persons**
--Compliance with National Laws and Regulations. --Conditions prior to project construction. --Status of landfill site. --Operation of landfill site. --The process and participation of the stakeholder consultation. --The impact of the project activity. --The complaint by local stakeholders and the implementation of the mitigation measures.	**Local environmental protection bureau:** Li Guocheng **Local residents:** Duan Haibo Yin Fengxia Mi Shenghua
-- Applicability of selected methodology -- Baseline of the project -- Emission factors -- Preparation of Joint PD -MR. --Compliance of the monitoring plan with the monitoring methodology --Compliance of monitoring with the monitoring plan --Assessment of data and calculation of GHG emission reductions	**Climate Bridge (Shanghai) Ltd. (Consultancy)** Ding Wenjun, Project manager

2.4 Site Visits

The joint validation and verification site visit was conducted on 04-March-2023. A ground inspection of the project was carried out on the project site and the village where the project is located. The on-site visits involved following activities,

An assessment of the implementation and operation of the project as per the Joint PD-MR /1/;

An inspection of the facilities including the Anaerobic animal manure management system, the Biogas pre-treatment system, the Organic fertilizer production plant, and the Biogas power generation system.

A review of information flows for generating, aggregating and reporting the monitoring parameters;

Interviews with relevant personnel to confirm that the operational and data collection procedures are implemented in accordance with the approved monitoring plan;

A cross-check between information provided in the Joint PD-MR /1/ and data from other sources such as operational records or similar data sources;

A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of VCS standard /5/ and the selected methodology /6//7/;

A review of calculations and assumptions made in determining the GHG data and emission reductions;

An identification of quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

2.5 Resolution of Findings

During the validation and verification of a project activity, to identify issues that need to be further elaborated upon, researched or added to in order to confirm that the project meets the VERRA's requirements and can achieve credible emission reductions, CCSC will issue Corrective Action Requests (CARs), Clarification Requests (CLs) or Forward Action Requests (FARs) depending on different situations.

The objective of this phase is to resolve issues related to the monitoring, implementation and operations of the project activity that could impair the capacity of the project activity to achieve emission reductions or influence the monitoring and reporting of emission reductions prior to CCSC's positive conclusion on the GHG emission reduction calculation.

Findings established during the validation and verification can either be seen as a non-fulfilment of criteria ensuring the proper implementation of a project or where a risk to deliver high quality emission reductions is identified.

A Corrective Action Request (CAR) will be raised if one of the following occurs:

a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the project participants, or if the evidence provided to prove conformity is insufficient;

b) Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the project participants;

c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;

d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the project participants.

A Clarification Request (CL) will be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

A Forward Action Request (FAR) will be raised, for actions if the monitoring and reporting require attention and/or adjustment for the next verification period.

There are 0 CL and 2 CARs that have been raised during the current validation and verification. A detailed list of the 2 CARs is included in Appendix C at the end of this report.

2.5.1 Forward Action Requests

This is the joint validation and 1st VCS verification, thus there is no FAR raised.

3 VALIDATION FINDINGS

3.1 Project Details

3.1.1 Project type, technologies and measures implemented, and eligibility of the project

- Type:

Through the onsite visit, interviewing the project developer and reviewing the Joint PD-MR/1/, it was confirmed that the project is to utilize animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters and the recovered biogas for electricity generation.

Therefore, it was confirmed that the sectoral scope is scope 01: Energy industries (renewable - / non renewable sources) and scope 13: Waste handling and disposal as per the UNFCCC Standard "Applicability of sectoral scopes version 01.0".

It was confirmed that the project is not a grouped project.

- Technologies and measures implemented:

The information and descriptions reported in section 1.1 of the VCS Joint-PD-MR/1//3/ have been checked.

The project involves the construction and operation of a centralized animal manure management system which collects manure waste (dairy cattle) from Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture in Weixian district. The project uses animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. The anaerobic digesters are expected to produce 13,760,000 m^3 biogas per year(Phase Ⅰ: 8,760,000 m^3; phase Ⅱ:5,000,000 m^3) according to the FSR/25/.

The project also involves the construction and operation of a biogas power plant. The recovered biogas is used for electricity generation at the biogas power plant. The running capacity of the project is 2.33MW (2 sets of 1.165MW CMM generators have been operated to generate electricity), while the intended installed capacity is 3.83MW (1.165MW * 2 + 1.5MW *1). The installation of the other 1.5MW will depend on the circumstances. The biogas power plant of the project is expected to supply 27,000 MWh of electricity to the North China Power Grid annually.

The technology is confirmed by site inspection and checking the FSR/25/, Contract and Annex for Construction of Anaerobic System/19/, Sales contract of container-type biogas cogeneration unit/20/, Technical Agreement for container-type biogas cogeneration unit Procurement /21/.

The project is expected to achieve an average annual emission reduction of 57,220 tCO_2e and a total emission reduction of 400,540 tCO_2e during the first 7-year crediting period.

In conclusion, it is verified that the summary description of the project in section 1.1 of the Joint PD-MR/3/ is in line with the Joint PD-MR template requirements and all the information has been provided and verified as correct.

• Eligibility of the project:

The PP has described and justified how the project is eligible under the scope of the VCS Program in Joint PD-MR/3/ as per the section 2.1.1 of VCS standard Version 4.4. The assessment is provided as below,

1. The project activity generates GHG emission reductions only including CO_2 which belong to the seven Kyoto Protocol greenhouse gases.

2. “The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction.” Via checking the Joint PD-MR of the project, it is verified that the project is implemented to utilize animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters and the recovered biogas for electricity generation, and the additionality is verified as actual, thus there is no assumption of having generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction for this project.

3. The VCS Program also excludes the following project activities under the circumstances indicated in Table 1 of section 2.1.1 of VCS Standard Version 4.4, via checking all the excluded project activities in the table, it is verified that this project type is not excluded by the scope of VCS program.

In conclusion, the project is eligible to the scope of the VCS Program.

3.1.2 Project design, including eligibility criteria for grouped projects

The project has been designed to be a single installation of an activity, not multiple project activity instances or as a grouped project.

3.1.3 Project proponent and other entities involved in the project

Through document review and on-site inspection, the assessment team confirmed the details of the project proponent as is shown below:

Organization name	Hebei Zhonghai Huaneng Energy Co., Ltd
Contact person	Ran Qiyan
Title	Manager
Address	Xiwang Village, Houguan Town, Weixian District, Xingtai City, Hebei Province, P.R China
Telephone	+86-021 23019950
Email	3542346576@qq.com

Organization name	Climate Bridge (Shanghai) Ltd.
Role in the project	Consultant
Contact person	Gao Zhiwen
Title	General Manager
Address	Block B, Level 24, Jiangong Mansion, 33 Fushan Road, Pudong New Area, Shanghai, China
Telephone	+86 021-62462036
Email	projects@climatebridge.com

3.1.4 Ownership

By checking business license /15/, Project Approval /16/ and the Construction Contract /19/, the assessment team confirmed that Hebei Zhonghai Huaneng Energy Co., Ltd is the project owner and also project proponent (PP) of the project. Therefore, the assessment team confirms that Hebei Zhonghai Huaneng Energy Co., Ltd has the legal right to control and operate the project.

3.1.5 Project start date

The project start date is 21-March-2021 which is verified as the date on which the project began generating GHG emission reductions or removals as per the concept in VCS Standard Version 4.4.

Via checking the monthly records during this monitoring period/23/, it is verified that 21-March-2021 is the date when all the system started fully operation and started gird-connected power generation, hence the start date is correct and in line with the VCS standard.

3.1.6 Project crediting period

The project adopts renewable crediting period of 7 years period.

Assessment team confirms that the crediting period is from 21-March-2021 to 20-March-2028 (both days included).

3.1.7 Project scale and estimated GHG emission reductions or removals

The total intended installed capacity of the project is 3.83 MW consisting of 2 sets of 1.165 MW and 1 set of 1.5MW generators. As the estimated annual average GHG emission reductions or removal per year is 57,220 tCO_2e which is less than 300,000 tonnes of CO_2e per year, thus the project falls in the category of Project. Therefore, the assessment team confirms that the project scale falls under Project.

Project Scale	
Project	✓
Large project	

Through checking emission reductions calculation spreadsheet provided by PP, the assessment team was able to confirm that the estimated GHG Emission Reductions or Removals of the project is as follows:

Year	Estimated GHG emission reductions or removals (tCO_2e)
21-Mar-2021~31-Dec-2021	44,835
01-Jan-2022~31-Dec-2022	57,220
01-Jan-2023~31-Dec-2023	57,220
01-Jan-2024~31-Dec-2024	57,220
01-Jan-2025~31-Dec-2025	57,220
01-Jan-2026~31-Dec-2026	57,220
01-Jan-2027~31-Dec-2027	57,220
01-Jan-2028~ 20-Mar-2028	12,385

Year	Estimated GHG emission reductions or removals (tCO_2e)
Total estimated ERs	400,540
Total number of crediting years	7
Average annual ERs	57,220

The above estimated emission reduction is confirmed by the assessment team via checking emission reduction calculation spreadsheet and relevant documents provided by the PP. The assessment team confirmed that the calculation is correct and conservative.

3.1.8 Project location

The project is located in Xiwang Village, Houguan Town, Weixian District, Xingtai City, Hebei Province, P.R China. The centre geographic coordinates of the project are east longitude 115° 26′20.24″ and north latitude 37°07′58.57″, which is confirmed during on-site inspection.

3.1.9 Conditions prior to project initiation

Via site interview with villagers, local government officer and checking the Environmental Impact Assessment (EIA) /26/ and the FSR/25/, CCSC confirmed that the scenario prior to the implementation of the proposed project activity is that electricity supplied from the NCPG connected fossil fuel fired power plants and the animal manure waste was left to decay in anaerobic manure management system (anaerobic lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility.

The baseline scenario is the same as the scenario existing prior to the implementation of the project activity, which has been confirmed during the site interview with local officer and checking the Environmental Impact Assessment (EIA)/26/ and the FSR/25/. Thus, it is verified that the baseline scenario is reasonable and correct.

Therefore, the project is confirmed not to be implemented to generate GHG emissions for the purpose of their subsequent reduction, removal or destruction.

3.1.10 Project compliance with applicable laws, statutes and other regulatory frameworks

The Joint PD-MR/1//3/ has stated that the project is in the field of renewable energy, this industry is confirmed as in line with the laws and regulations as listed in the Joint PD-MR.

Also, the project is legally approved by local government by checking the project approval/16/, Grid-connected dispatching protocol /27/ and Environmental Impact Assessment (EIA) approval /26/, which means that the project is in line with the national and local laws and regulations based on the local expertise of the VVB. And as stated in the Joint PD-MR /1//3/ and confirmed by interviewing with local officer, CCSC verified that the project will be subject to regular inspection by local government during the implementation and operation period to ensure continuous compliance.

In conclusion, the project complies with all relevant local, regional and national laws, statutes and regulatory frameworks in China.

3.1.11 Participation under other GHG programs:

-Projects registered (or seeking registration) under other GHG program(s)

The project has neither been registered nor seeking registration under any other GHG programs which has been confirmed via checking the UNFCCC, GS, VCS and other GHG schemes' website. It is verified that the project is seeking registration only in VCS program.

-Rejection by other GHG programs

The project has neither been rejected by any other GHG programs which has been confirmed via checking the UNFCCC, GS, VCS and other GHG schemes' website.

3.1.12 Other forms of credit and supply chain (Scope 3) emissions:

-Emissions trading programs and other binding limits

The project proponent is not part of any emission trading program which has been confirmed via checking the UNFCCC, GS, VCS and other GHG schemes' website. The net GHG emission reductions during this monitoring period from the project was not used for compliance with emission trading programs or to meet binding limits on GHG emissions. The project activity has not participated under any other GHG programs which has been confirmed via checking the UNFCCC, GS, VCS and other GHG schemes' website, it is verified that the implementation of the project is consistent with the description in the Joint PD-MR and the monitoring system is fully functional to generate Verified Carbon Units (VCUs) without any double counting for this monitoring period from 21-March-2021 to 30-Septemper-2022.

Besides, based on VVB's local expertise, China has a national emissions trading scheme only cover the high-emission industries, such as power generation sector that emitted at least 26,000 tons of CO_2e/year which has been verified in the public website, and it is confirmed that the project activity is not included the mandatory emission control scheme and there is no emission cap enforced for the project owner by checking the enforced company list in public information. Hence, it is confirmed that the emission reductions will not be double counted.

According to the Verra Releases Updates to VCS Program updated on 21-December-2022, issuance of public statement(s) to help prevent Scope 3 emissions double claiming and Email notification of the potential risk of Scope 3 emissions double claiming are not required in the Joint PD-MR of this Project.

-Other forms of environmental credit sought or received and eligible to be sought or received

The Project has no intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

Renewable energy certificates are available for trading in the host country. However, the same is not availed by the PP. The undertaking regarding the same is submitted by PP which is acceptable to the assessment team. The assessment team also checked the REC website and found the declaration to be correct.

3.1.13 Additional information relevant to the project, including:

-Leakage management for AFOLU projects

Not applicable.

-Commercially sensitive information

No commercially sensitive information has been excluded from the public version of the project description.

-Sustainable development contributions

Via on-site inspection, checking the evidence provided as below and interview with the project implementer, CCSC verified that the project contributes the sustainable development through the following aspects:

- Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all (Goal 8 of UN SDG)-verified by interview with staffs and checking the Social security certificate of employees/34/. It is confirmed that 19 job opportunities have been provided during this monitoring period, hence the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable;
- Take urgent action to combat climate change and its impacts (Goal 13 of UN SDG,) – verified by checking the Emission Reduction Calculation sheet/2/, it is confirmed that total 47,385 tCO_2e Emission Reduction have been achieved during this monitoring period, hence the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable.
- Ensure access to affordable, reliable, sustainable and modern energy for all (Goal 7 of UN SDG,) – verified by checking the Electricity transaction note issued by the grid company during this monitoring period/24/ and Monthly records during this monitoring period/23/, it is confirmed that 17,296.840 MWh of electricity from renewable sources has been supplied to the power grid during the monitoring period from 21-March-2021 to 30-September-2022.

Based on above assessment and demonstration, it is concluded that the description in the project description is accurate, complete, and the nature of the project is actual and understandable, and it is confirmed that the project has been implemented as described in the project description via site inspection.

CAR 1 and CAR 2 were raised and successfully closed. Refer to Appendix 4 for detail assessment.

3.2 Participation under Other GHG Programs

The project has neither been registered nor seeking registration under any other GHG programs which has been confirmed via checking the UNFCCC, GS, VCS and other GHG schemes' website.

It is verified that the project is seeking registration only in VCS program.

3.3 Safeguards

3.3.1 No Net Harm

The Environmental Impact Assessment (EIA)/26/ Report of the project activity was compiled by Hebei Shihuan Environmental Assessment Service Co., Ltd. and approved by Weixian Branch of Xingtai Ecological and Environmental Bureau on 04-December-2019, with approval No. "Xinghuanweishen [2019] No.79"/17/. The assessment team confirms all environmental impacts has been analysed and no net harm was detected. Refer to section 3.3.3 of this report for detailed environmental impacts arising from the Project construction and operation phase.

Through interview, it is confirmed by the assessment team that the implementation of the project will improve local socio-economic development and contribute to the sustainable development as described in section 3.2 of this report above.

In conclusion, it is confirmed by the assessment team that the project has no negative impacts on local environment and socio-economy. And no net harm on local environment and social community has been detected for the project.

3.3.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to perform the relevant stakeholders consultation, prior to the construction process. The assessment team checked the relevant dates during the validation and confirms that the project proponent uses the following methods to communicate with stakeholders:

PP informed residents around the project area and government officials the general information about the project and carried out questionnaire survey by putting up public announcements and phone calls. Then local stakeholder consultation was conducted through distributing questionnaire survey /29/ to local stakeholders in March 2021 by the project owner. The survey was conducted through distributing and collecting responses of questionnaires at a variety of places. The questionnaire was reasonably designed to assess the project impacts on the local environment and social economic development. In total, 30 out of 30 questionnaires were returned with a 100% response rate.

Furthermore, through on-site inspection, the assessment team confirmed that communications with stakeholders were conducted. Key implementation schedules or changes of the project has been and will be communicated to the local authority, who will inform the residents. The comments and suggestions from residents will be collected by the local authority and get back to project owner. Local government agencies and authorities will conduct spot checks on the project implementation on a regular basis, and give suggestions on potential issues.

By checking the questionnaire survey/29/, the assessment team confirms that the local stakeholder has no negative comments for the construction of the project activity. Meanwhile, by checking the filled questionnaires from local stakeholders, the assessment team confirmed that communications with local stakeholders was being carried out in March 2021 by the project owner during this monitoring period. There are no negative comments received for the project during this monitoring period.

By checking the questionnaire survey/29/, during this monitoring period, there were also 30 questionnaires distributed on 22-Jun-2022 and according to the result there were no negative comments received. Also, the project proponent received no comments for the project by phone call. Furthermore, via checking questionnaires from local stakeholders and interviewing with the residents and government officials during the on-site inspection, it is confirmed by the assessment team that during the implementation stage of this monitoring period, local authority has conducted follow-up interviews and ongoing communication with local stakeholders to collect any comments or inputs to the project implementation at periodic intervals. It is confirmed by the assessment team that there were no negative comments and issues from the local stakeholders during this monitoring period and the project passed all the periodic spot checks by local government.

The mechanism for ongoing communication with local stakeholders

Via above assessment, it is confirmed that not only the communications with local stakeholders took place regularly during the project implementation but also local stakeholders can raise their concerns and opinions directly by making a phone call to the project proponent. Furthermore, PP has informed the local authorities of key implementation events or changes of the project and local government agencies is in charge of supervising the project implementation on a regular basis which has been confirmed by site interview with local officers.

Therefore, the assessment team was able to confirm that the stakeholder consultation was adequate and appropriate.

3.3.3 Environmental Impact

An Environmental Impact Assessment has been conducted by the project participants. The assessment team has reviewed the documentation of the presented information, e.g., EIA Report /26/ and EIA Approval /17/. The EIA Approval confirms the correctness of the approach used by PP. In conclusion, the PP has followed the requirements of the host country with regards to addressing environmental impacts as below:

Air pollution

By checking EIA Report /26/ and site interview with stakeholders, the assessment team confirmed that construction dust and road dust during the construction had a certain effect to the surrounding areas of construction site.These adverse effects are accidental, temporary and partial, as long as to take effective measures and strengthen management, the scope of its influence is generally limited to the surrounding area of the construction site and will disappear along with the end of construction.

By checking EIA Report /26/ and on-site inspection, the assessment team confirmed that air pollution during the operation mainly includes the exhaust gas of the biogas generator and the flare, and dust from organic fertilizer plant. The exhaust gas is to be treated through 2-stage purification equipment before being discharged through exhaust cylinders with 15m high. The dust from organic fertilizer plant is to be filtered by dust collector and reused in the organic fertilizer plant. The NOx, sulfur dioxide, H_2S, NH_3 and particulate matter in all the exhaust gas by the project meets the Table 2 standard limit of the "Odorous Pollutant Emission Standard" (GB14554-93).

Wastewater

During the construction period, wastewater mainly includes both construction and domestic water use. By checking EIA Report /26/ and site interview with stakeholders, the assessment team confirmed that the construction wastewater was been controlled by construct temporary diversion ditches on the construction site, set up a sedimentation tank, and reuse the washing water for construction machinery as much as possible after simple treatment of equipment and vehicles. Domestic sewage was been collected and processed by setting up mobile toilets, and regularly handed over to the sanitation department to remove garbage and excrement, which can effectively prevent the sewage generated by construction workers from polluting the water environment.

During the operation period, the wastewater is mainly domestic sewage and biogas slurry from anaerobic digesters. By checking the EIA Report /26/ and on-site inspection, the assessment team confirmed that the domestic sewage of the project was pretreated in a septic tank and reused in anaerobic digesters. Most biogas slurry was also be recycled in anaerobic digesters, surplus biogas slurry was used as material for organic fertilizer production. No wastewater was discharged to the environment. and satisfied with relevant national standard.

Noise

By checking the EIA Report /26/ and EIA approval/17/, the assessment team confirmed that the noise generated by the project construction has a slight impact on the surrounding environment. Reasonable measures have been taken to reduce the noises to acceptable range. The noise control met the requirements in the national standard Emission Standard of Environment Noise for Boundary of Construction Site (GB 12523-2011).

As per the EIA Report /26/, EIA approval/17/and on-site inspection, the assessment team confirmed that during the operation period, the high noise equipment of the project mainly consists of various fans and pumps. After sound insulation, vibration reduction and noise elimination, the noise at the plant boundary can meet the requirements of Class II standard of the "Emission Standard for Industrial Enterprises Noise at Boundary" (GB12348-2008) after greening and distance attenuation.

Solid Waste

By checking the EIA Report /26/, EIA approval/17/, and interview with stakeholders, the assessment team confirmed during the construction period, a certain amount of construction waste and domestic waste from construction workers will be generated, of which part of the construction waste will be recycled, and the unusable waste will be handed over to the sanitation department for disposal.

During the operation period, the solid waste of the project mainly includes biological sludge of desulfurization tower, sludge of biogas purification system, digestate, laboratory waste reagents and waste liquid, and domestic waste of employees. By checking the EIA Report /26/ and on-site inspection, the assessment team confirmed that the following measures were taken:

1) Biological sludge of desulfurization tower

After pressure filtration, it is temporarily stored in the desulfurization tank and regularly sold to the sulfur manufacturer.

2）Sludge of biogas purification system

After dewatering by the dehydrator in the factory area, it will be collected and cleaned and disposed by the local sanitation department.

3) Digestate

The digestate produced after anaerobic fermentation is separated by solid-liquid separation, and the solid part is squeezed twice before entering the drying drum. The dried part is used as cow bedding material, and the liquid part is treated with aerobic aeration before being returned to farmland for fertilizer. The liquid part will be sent to aerobic treatment as soon as digestate is separated by solid-liquid separation. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve anaerobic composting of digestate. Hence proper conditions and procedures (not resulting in methane emissions) for digestate is ensured.

4) Laboratory waste reagents and waste liquid

Laboratory waste reagents and waste liquid will be properly collected and stored in the hazardous waste temporary deposit, and will be submitted to the relevant qualified entities for centralized treatment regularly.

5）Domestic waste of employees

It was be collected and cleaned and disposed by the local sanitation department.

In conclusion, after the above measures are performed, the negative impacts on environment will be minimized below the requirements of laws and regulations during the construction and operational period.

3.3.4 Public Comments

The assessment team noted that this project was open for public comment from 19-December-2022 to 18-January-2023. The detail was checked by the assessment team in the following web platform:

https://registry.verra.org/app/projectDetail/VCS/3984

During the period, no public comments were received.

3.3.5 AFOLU-Specific Safeguards

The project is not AFOLU project, and thus this section is not required.

3.4 Application of Methodology

3.4.1 Title and Reference

The assessment team checked that following methodology and tools are applicable for the project activity as below:

AMS-III.D Methane recovery in animal manure management systems (version 21.0);

AMS-I. D Grid connected renewable electricity generation (version 18.0);

Project and leakage emissions from anaerobic digesters (version 02.0);

Tool to calculate the emission factor for an electricity system (version 07.0);

Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (version 03.0)

3.4.2 Applicability

The assessment team has, by means of an on-site inspection and document review, assessed that the project activity meets the applicability of methodology and related tools.

Table 3-1 Assessment of Applicability conditions of AMS-III.D (version21.0).

No.	Applicability conditions	Justifications	Validation Opinion
1	This methodology covers project activities involving the replacement or modification of anaerobic animal manure management systems in livestock farms to achieve methane recovery and destruction by flaring/combustion or gainful use of the recovered methane. It also covers treatment of manure collected from several farms in a centralized plant	Applicable. The project is a centralized plant treats animal manure waste collected from Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture.The project replaces existing anaerobic animal manure anagement systems (open lagoon) in livestock farms to achieve methane recovery and destruction by flaring/combustion or gainful use of the recovered methane.	By checking the FSR/25/ and site visit, it is able to confirm: the project involves the construction and operation of the anaerobic animal manure management system, the biogas pre-treatment system, the organic fertilizer production plant and the biogas power generation system. Therefore, this criterion is applicable.
2	This methodology is only applicable under the following conditions: a) The livestock population in the farm is managed under confined conditions	Applicable. All the livestock population in the farms within the project boundary is managed under confined conditions	By checking the FSR, the dynamic table of cattle and site visit, it is able to confirm that the livestock population in the farm is managed under confined conditions Therefore, this criterion is applicable.

	b) Manure or the streams obtained after treatment are not discharged into natural water resources (e.g., river or estuaries), otherwise "AMS-III.H Methane recovery in wastewater treatment" shall be applied	Applicable. As per the FSR, manure or the streams obtained after treatment are not discharged into natural water resources (e.g., river or estuaries)	By checking the FSR, interviewing with the officer from local Environmental Bureau and on-site inspection, it is able to confirm: Manure or the streams obtained after treatment are not discharged into natural water resources Therefore, this criterion is applicable.
	c) The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C	Applicable. The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C. The annual average temperature at the project site is 13.1℃.	By checking the FSR, and interviewing the local environmental officer during site visit, it is able to confirm: The annual average temperature at the project site is 13.1℃ Therefore, this criterion is applicable.
	d) In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m	Applicable. In the baseline scenario the retention time of manure waste in the anaerobic lagoons is greater than one month,	By interviewing with operating staff of the project during site inspection, it is able to confirm in the baseline scenario, the retention time of manure waste in the

		and their depths are at least 1 m.	anaerobic lagoons is greater than one month, and their depths are at least 1 m. Therefore, this criterion is applicable.
	e) No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario	Applicable. No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario.	By checking the FSR and site visit, it is able to confirm that there is no methane recovery and destruction by flaring or combustion in the baseline scenario Therefore, this criterion is applicable.
3	The project activity shall satisfy the following conditions: (a) The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of "AMS-III.AO Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions and procedures (not resulting in methane emissions) must be ensured;	Applicable. The residual waste from the animal manure management system of the project will be used to produce organic fertilizer, which is handled aerobically and will not result in methane emissions.	By checking the FSR and site visit, it is able to confirm the organic fertilizer production plant has been constructed for producing organic fertilizer, which the digestate produced after anaerobic fermentation is separated by solid-liquid separation, and the solid part is squeezed twice before entering the drying drum. The dried part is used as cow bedding material, and the liquid part is treated with aerobic aeration before being returned to farmland for fertilizer. The liquid part will be sent to aerobic treatment as soon as digestate is separated by solid-liquid separation. Digestate will not be

			stored under anaerobic conditions. In addition, the project does not involve anaerobic composting of digestate. Therefore, leakage emissions of the project associated with the anaerobic digester is not accounted for, leakage emission is 0. Therefore, this criterion is applicable.
	(b) Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared;	Applicable. Biogas produced by the project are used directly to produce electricity, the emergency flare ensure biogas would be destroyed while exigencies happened	By checking the FSR, the Contract and annex for construction of Anaerobic System/19/ and site visit, it is able to confirm the biogas produced by the digester is used or flared Therefore, this criterion is applicable.
	(c) The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when removed from the animal barns is larger than 20%, this time constraint will not apply.	Applicable. The storage time of the manure after removal from the animal barns, including transportation will not exceed 45 days before being fed into the anaerobic digester.	By checking the FSR and site visit, it is able to confirm that the storage time of the manure after removal from the animal barns, including transportation will not exceed 45 days before being fed into the anaerobic digester. Therefore, this criterion is applicable.

4	Projects that recover methane from landfills shall use "AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure shall use "AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology "AMS-III.AO Methane recovery through controlled anaerobic digestion".	Irrelevant. The project does not involve landfill methane recovery, wastewater treatment, composting animal manure, or co-digestion of animal manure and other organic matters.	By checking the FSR and site visit, it is able to confirm that the project does not involve landfill methane recovery, wastewater treatment, composting animal manure, or co-digestion of animal manure and other organic matters. Therefore, this criterion is irrelevant.
5	Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the project activity, it should be taken into account accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component.	Applicable. Part of the recovered biogas is used to produce and deliver electricity to the power grid, therefore AMS.I. D is applied. Part of the biogas is used to power auxiliary equipment of the project activity, zero is used as its emission factor, and this part of biogas is not eligible as an SSC CDM Type I project component, which means no emission reductions will be	By checking the FSR, power purchase contract and site visit, it is able to confirm that part of the recovered biogas is used to produce and deliver electricity to the grid, and part of the biogas is used to power auxiliary equipment of the project activity, hence the emission factor for the 2nd part of biogas utilization is zero and no emission reductions will be claimed for it. Therefore, this criterion is applicable.

		claimed for this part of biogas utilization.	
6	New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the "General guidelines for SSC CDM methodologies".	Applicable. The project is a Greenfield project. The emission reduction sourced from methane recovery is 44,184 tCO_2e/yr, which is lower than the threshold of 60,000 tCO_2e/yr; the total installed capacity of the project is 3.83MW, which is lower than the threshold of 15MW. Therefore, the project is in line with "General Guidelines to SSC CDM methodologies"	By checking the ER calculation spreadsheet during the 1st monitoring period and site visit, it is able to confirm that the project complies with the related and relevant requirements in the "General guidelines for SSC CDM methodologies".
7	The requirements concerning demonstration of the remaining lifetime of the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies".	The project is a Greenfield project, and does not involve the replaced equipment; therefore, this is irrelevant.	By checking the FSR and site visit, it is able to confirm that no equipment needs to be replaced. Therefore, this criterion is irrelevant.
8	Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO_2 equivalent annually from all Type III components of the project activity.	Applicable. The emission reductions from the recovery and destruction of	By checking the ER calculation spreadsheet and site visit, it is able to confirm that the

		methane (viz. Type III components of the project) are 44,184 tCO_2e/yr, which is less than 60 kt CO_2 equivalent.	emission reductions from the recovery and destruction of methane (viz. Type III components of the project) are 44,184 tCO_2e/yr, which is less than 60 kt CO_2 equivalent.
Conclusion: Methodology AMS-III.D (version21.0). is applicable to the project activity.			

Table 3-2 Assessment of Applicability conditions of AMS-I.D (version18.0).

No.	Applicability conditions	Justifications	Validation Opinion
1	This methodology comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass: (a) Supplying electricity to a national or a regional grid; or (b) Supplying electricity to an identified consumer facility via national/regional grid through a contractual arrangement such as wheeling.	Applicable. The project will use biogas to generate electricity, and the electricity will be supplied to North China Power Grid.	By checking the FSR, Grid-connected dispatching protocol /27/and site visit, it is able to confirm the project uses biogas to generate electricity, and the electricity is supplied to NCPG.
2	Illustration of respective situations under which each of the methodology (i.e., “AMS-I.D.: Grid connected renewable electricity generation”, “AMS-I.F.: Renewable electricity generation for captive use and mini-grid” and “AMS-I.A.: Electricity generation by the user) applies is included in the appendix.	The illustration of respective situations for AMS-I.D stated in the appendix is the same as applicability condition No. 1 above. No need for further justification.	same as No.1 in this table
3	This methodology is applicable to project activities that:	Applicable. The project is a	By checking the FSR and site visit, it is able to confirm the project is a greenfield plant.

	(a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant(s); (c) Involve a retrofit of (an) existing plant(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s).	greenfield plant.	
4	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology: (a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m2; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m2.	Irrelevant. The project is not a hydro power plant.	By checking the FSR and site visit, it is able to confirm the project is not a hydro power plant.
5	If the new unit has both renewable and non-renewable components (e.g., a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW.	Irrelevant. The project does not involve non-renewable components.	By checking the FSR and site visit, it is able to confirm the project does not involve non-renewable components.
6	Combined heat and power (co-generation) systems are not eligible under this category.	According to the FSR, the only output for the project is electricity and the project does not involve	By checking the FSR and site visit, it is able to confirm the project does not involve co-generation system.

		co-generation system.	
7	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct1 from the existing units.	Irrelevant. The project does not involve capacity addition.	By checking the FSR and site visit, it is able to confirm the project does not involve capacity addition.
8	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW.	Irrelevant. The project does not involve retrofit, rehabilitation or replacement.	By checking the FSR and site visit, it is able to confirm the project does not involve retrofit, rehabilitation or replacement.
9	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid, then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as "AMS-I.C.: Thermal energy production with or without electricity" shall be explored.	Irrelevant. The project does not involve landfill gas, waste gas, wastewater treatment and agro-industries projects.	By checking the FSR and site visit, it is able to confirm the project does not involve landfill gas, waste gas, wastewater treatment and agro-industries projects.
10	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass" shall apply.	Irrelevant. The project does not involve biomass.	By checking the FSR and site visit, it is able to confirm the project does not involve biomass.

In addition, the project meets the applicability conditions of the applied tools as follows:

	Tool to calculate the emission factor for an electricity system (version 07.0)		
No.	**Applicability conditions**	**Justifications**	**Validation Opinion**
1	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a	Applicable. The project utilizes	By check the FSR, the ER calculation

	project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects)	biogas for power generation to supply electricity to NCPG that dominated by fossil-fuel plants. Therefore, the tool is applied to estimate the OM, BM and/or CM when calculating baseline emissions for the project.	spreadsheet and site visit, it is able to confirm that the project utilizes biogas for power generation to supply electricity to NCPG that dominated by fossil-fuel plants. Therefore, the tool is applied to estimate the OM, BM and/or CM when calculating baseline emissions for the project.
2	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option II a and option II b. If option II a is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	In the host country as off-grid power generation is not significant. Therefore, emission factor for the project electricity system is calculated only for the grid power plants. Thus, this applicability criterion is satisfied.	According to the situation in the host country and by checking the ER calculation spreadsheet and site visit, the emission factor for the project electricity system can be calculated only for grid power plants.
3	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Applicable. The project electricity system is located in a non-Annex I country.	By checking the Annex I country and site visit, it is able to confirm the project is not located in an Annex I country.

4	Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.	The project is not involved in biofuels.	By checking the FSR and site visit, it is able to confirm that the project does not involve biofuels.
	Project and leakage emissions from anaerobic digesters (version 02.0)		
1	The following sources of project emissions are accounted for in this tool: (a) CO_2 emissions from consumption of electricity associated with the operation of the anaerobic digester; (b) CO_2 emissions from consumption of fossil fuels associated with the operation of the anaerobic digester; (c) CH_4 emissions from the digester (emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester); and (d) CH_4 emissions from flaring of biogas.	Applicable. All sources of project emissions have been accounted.	By checking the ER calculate sheet and the Joint PD-MR, it is able to confirm that all sources of project emissions have been accounted.
2	The following sources of leakage emissions are accounted for in this tool: (a) CH_4 and N_2O emission from composting of digestate; (b) CH_4 emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond.	Irrelevant. The project does not involve composting, anaerobic decay or anaerobic storage of digestate.	By checking the FSR and site visit, it is able to confirm that the project does not involve composting, anaerobic decay or anaerobic storage of digestate.
3	Emission sources associated with N_2O emissions from physical leakages from the digester, transportation of feed material and digestate or any other on-site transportation, piped distribution of the biogas, aerobic treatment of liquid digestate and land application of the digestate are neglected because these are minor emission sources or because they are accounted in the methodologies	Applicable. As per the applied methodology, N_2O emissions are neglected because these are minor emission sources.	By checking the FSR, the applied methodology, and site visit, it is able to confirm the N_2O emissions are neglected because these are minor

	referring to this tool.		emission sources.
	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)		
	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.	Applicable. The source of electricity consumption of the project is Scenario A: Electricity consumption from the grid	By site visit, it is able to confirm that scenario A: electricity consumption from the grid is applicable to the source of electricity consumption of the project.
	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to	Applicable. The electricity generated by the project is supplied to the NCPG (Scenario I).	By site visit, it is able to confirm the electricity generated by the project is supplied to the NCPG, which is in line with Scenario I.

	consumers/electricity consuming facilities; o©(c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.		
	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO_2 emissions.	Applicable. No captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage.	By site visit, it is able to confirm no captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage.

3.4.3 Project Boundary

The project boundary basically defines the physical and geographical boundary of the project facility and it is well defined in the Joint PD-MR/3/ (section 3.3) according to AMS-III.D Methane recovery in animal manure management systems (version 21.0)/53/ and AMS-I. D Grid connected renewable electricity generation (version 18.0)/54/. The project boundary includes followings,

(I) The physical and geographical sites of the livestock;

(II) The centralized animal manure management system;

(III) The project power plant and all power plants physically connected to the NCPG

Via site inspection and checking the FSR/5/, it is verified that project boundary is clearly defined in the Joint PD-MR/3/ as per the methodology/6//7/.

Emissions sources included in the project boundary have been appropriately included in the Joint PD-MR/1//3/.

The sources and GHG gases involved for the Project activity are as below:

Source		Gas	Included?	Justification/Explanation
Baseline	Direct emissions from the manure treatment	CO_2	No	Excluded for simplification.
		CH_4	Yes	The major source of emissions in the baseline
		N_2O	No	Excluded for simplification.

Source		Gas	Included?	Justification/Explanation
	processes			
	Emissions from electricity consumption /Generation	CO_2	Yes	The major source of emissions in the baseline
		CH_4	No	Excluded for simplification.
		N_2O	No	Excluded for simplification.
Project	Physical leakage of biogas in the manure management systems	CO_2	No	Excluded for simplification.
		CH_4	Yes	Main emission source.
		N_2O	No	Excluded for simplification.
	Emissions from flaring or combustion of the gas stream	CO_2	No	Excluded for simplification.
		CH_4	No	Maybe a main emission resource[1].
		N_2O	No	Excluded for simplification.
	Emissions from use of fossil fuels or electricity	CO_2	Yes	The project consumes electricity during operation, so emission from use of electricity is the main emission source. The project does not involve fossil fuel consumption, so the emission from use of fossil fuels is not included.
		CH_4	No	Excluded for simplification.
		N_2O	No	Excluded for simplification.
	Emissions from incremental transportation distances	CO_2	No	Excluded for simplification.
		CH_4	No	Excluded for simplification.
		N_2O	No	Excluded for simplification.
	Emissions from the storage of manure	CO_2	No	Excluded for simplification.
		CH_4	No	By checking the FSR and interview with PP during site visit, the storage time of the manure after removal from the animal barns, including transportation, is within 24 hours before being fed into the anaerobic digester, hence Emissions from the storage of manure is not accounted for.

[1] Only when emergency occurs, the emergency flare system will be used and the biogas will be flared. In this situation, there will be project emission from flaring.

Source		Gas	Included?	Justification/Explanation
		N_2O	No	Excluded for simplification.

It is concluded that the project boundary and selected sources are correctly justified for the project.

3.4.4 Baseline Scenario

As per para. 17 of AMS-III.D, the baseline scenario is the situation where, in the absence of the project activity, animal manure is left to decay anaerobically within the project boundary and methane is emitted to the atmosphere.

As per para. 19 of AMS-I. D, the baseline scenario is that the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid.

The assessment team confirms that the Joint PD-MR conforms to the guidance given by EB via CDM Validation and Verification Standard for project activities version 03.0 and Verra via VCS standard version 4.4.

The project uses animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. The recovered biogas will be used for electricity generation at the project site. The electricity generated by the project will be delivered to North China Power Grid (NCPG). In the absence of the project, the animal manure waste was left to decay in anaerobic manure management system (open lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility. Equivalent amount of electricity was supplied from the NCPG connected fossil fuel fired power plants.

The assessment team confirms that the Project has been approved by Chinese government by checking the Project approval /16/ and Environmental Impact Assessment (EIA) approval/17/. It is concluded that the project is in complicate with all laws and regulations in China.

3.4.5 Additionality

As per para. 15-16 of applied methodology AMS-III.D, project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site, that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the “ Guidelines on the demonstration of additionality of small-scale project activities”. This additionality condition also applies to greenfield project activities. Furthermore, for project activities applying this methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component.

The regulations relative to the project in China are identified as below.

I) Law of the People's Republic of China on Environment Protection;

II) Law of the People's Republic of China on the Prevention and Control of Solid Waste Pollution;

III) Law of the People's Republic of China on the Prevention and Control of Atmospheric Pollution;

IV) Regulations on prevention and control of pollution from large scale livestock and poultry breeding;

V) Discharge standard of pollutants for livestock and poultry breeding (GB 18596-2001);

VI) Technical standard of pollution prevention for livestock and poultry breeding (HJ/T 81-2001).

The project is to build a centralized anaerobic animal manure treatment system which collects manure waste (dairy cattle) from Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture in Weixian district. The project uses animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters. The recovered biogas is used for electricity generation at the project site. Prior to the implementation of the project, the animal manure waste was left to decay in anaerobic manure management system (open lagoon) at the livestock farms and methane is emitted to the atmosphere directly without any methane recovery and destruction facility. Equivalent amount of electricity was supplied from the NCPG connected fossil fuel fired power plants. By interviewing the project owner and employees, it is confirmed by the assessment team that only the project collects manure waste from Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture in Weixian district to recover biogas and generate electricity. By checking Project Approval and EIA Approval, the total intended installed capacity of the project activity is 3.83 MW(2*1.165MW+1*1.5MW), which is below 5 MW. Therefore, the project is deemed automatically additional.

Furthermore, by checking applicable regulations, the assessment team confirm that there are no laws or regulations that require the collection and destruction of methane from livestock manure. Thus, the project activity is not mandatory by any law, statute or other regulatory framework. Therefore, it is concluded that the project fulfils the VCS requirements regarding regulatory surplus.

3.4.6 Quantification of GHG Emission Reductions and Removals

The assessment team checked the baseline, project and leakage calculation and confirmed that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

3.4.6.1 Baseline Emissions

a) Methane emissions from the manure treatment processes in the absence of the project activity;

b) Electricity generation using fossil fuels or supplied by the grid in the absence of the project activity;

$$BE_y = BE_{CH4,y} + BE_{Elec,y} \quad (1)$$

Where:

BE_y	=	Baseline emissions in year *y* (t CO_2e/yr)
$BE_{CH4,y}$	=	Baseline emissions from the manure treatment processes in year *y* (t CO_2e/yr)
$BE_{Elec,y}$	=	Baseline emissions associated with electricity generation in year *y* (t CO_2/yr)

Via reviewing the Joint PD-MR/3/, it is confirmed the ERs is calculated as per the applied methodology (AMS-III.D ver. 21.0 and AMS-I. D ver.18.0) with follow steps listed below.

Baseline Emissions from the manure treatment processes in year y (*$BE_{CH4,y}$*)

Via checking the applied methodology (AMS-III.D ver. 21.0), there are two options for the baseline emissions calculation:

(a) Using the amount of the waste or raw material that would decay anaerobically in the absence of the project activity, with the most recent IPCC Tier 2 approach (please refer to the chapter 'Emissions from Livestock and Manure Management' under the volume 'Agriculture, Forestry and other Land use' of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories). For this calculation, information about the characteristics of the manure and of the management systems in the baseline is required. Manure characteristics include the amount of volatile solids (VS) produced by the livestock and the maximum amount of methane that can be potentially produced from that manure (B_o);

(b) Using the amount of manure that would decay anaerobically in the absence of the project activity based on direct measurement of the quantity of manure treated together with its specific volatile solids (SVS) content.

The project applies option (a) to calculate baseline emissions from the manure management ($BE_{CH4,y}$), and baseline emissions are determined as follows:

$$BE_{CH4,y} = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bl,j} \quad (2)$$

Where:

GWP_{CH4}	=	Global Warming Potential (GWP) of CH_4 applicable to the crediting period (tCO_2e/tCH_4)
D_{CH4}	=	CH_4 density (0. 67kg/m^3 at room temperature (20ºC) and 1 atm pressure)
UF_b	=	Model correction factor to account for model uncertainties (0.94)
LT	=	Index for all types of livestock
j	=	Index for animal manure management system
MCF_j	=	Annual methane conversion factor (MCF) for the baseline animal manure management system j
$B_{0,LT}$	=	Maximum methane producing potential of the volatile solid generated for animal type LT (m^3CH_4/kg-dm)
$N_{LT,y}$	=	Annual average number of animals of type LT in year y (numbers)
$VS_{LT,y}$	=	Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)
$MS\%_{Bl,j}$	=	Fraction of manure handled in baseline animal manure management system j

a) The maximum methane-producing capacity of the manure ($B_{o,LT}$) varies by species and diet. The preferred method to obtain $B_{o,LT}$ measurement values is to use data from country-specific published sources, measured with a standardised method ($B_{o,LT}$ shall be based on total as-excreted VS). These values shall be compared to IPCC default values and any significant differences shall be explained. If country specific B_o values are not available, default values from tables 10 A-4 to 10 A-9 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10 can be used, provided that the project participants assess the suitability of those data to the specific situation of the treatment site.

b) VS are the organic material in livestock manure and consist of both biodegradable and non-biodegradable fractions. For the calculations the total VS excreted by each animal species is required.

   If country specific VS values are not available, IPCC default values from 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 table 10 A-4 to 10 A-9 can be used provided that the project participants assess the suitability of those data to the specific situation of the treatment site particularly with reference to feed intake levels;

c) $B_{o,LT}$ values or VS values applicable to developed countries can be used provided the following four conditions are satisfied:

i. The genetic source of the livestock originates from an Annex I Party;

ii. The farm uses formulated feed rations (FFR) which are optimized for the various animal(s), stage of growth, category, weight gain/productivity and/or genetics;

iii. The use of FFR can be validated (through on-farm record keeping, feed supplier, etc.);

iv. The project specific animal weights are more similar to developed country IPCC default values.

d) Methane Conversion Factors (MCF) values are determined for a specific manure management system and represent the degree to which B_o is achieved. Where available country-specific MCF values that reflect the specific management systems used in particular countries or regions shall be used. Alternatively, the IPCC default values provided in table 10.17 of 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 can be used. The site annual average temperature is taken from official data at the nearest meteorological station, or from data available from historical on-site observations.

e) The annual average number of animals ($N_{LT,y}$) is determined as follows:

$$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365}) \quad (3)$$

Where:

$N_{da,y}$ = Number of days animal is alive in the farm in the year y (numbers)

$N_{p,y}$ = Number of animals produced annually of type LT for the year y (numbers)

Baseline emissions associated with electricity generation in year y (*$BE_{elec,y}$*)

Via checking the applied methodology AMS-I.D ver. 18.0 , baseline emissions associated with electricity generation in year y is calculated according to equation (4) as below:

$$BE_{elec,y} = EG_{PJ,y} \times EF_{grid,y} \quad (4)$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh)

$EF_{grid,y}$	=	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (t CO_2/MWh)

Calculation of combined margin CO_2 emission factor for grid ($EF_{grid,y}$)

According to "Tool to calculate the emission factor for an electricity system" (version 07.0). The following six steps are applied to determine OM, BM, and CM used for calculating the project baseline emissions:

- **Step 1:** Identify the relevant electricity systems;
- **Step 2:** Choose whether to include off-grid power plants in the project electricity system (optional);
- **Step 3:** Select a method to determine the operating margin (OM);
- **Step 4:** Calculate the operating margin emission factor according to the selected method;
- **Step 5:** Calculate the build margin (BM) emission factor;
- **Step 6:** Calculate the combined margin (CM) emission factor.

As China DNA has published the calculation method for emission factor of grid, the published data and method have been applied for this project to calculate operating margin (OM) and build margin, as following steps:

Step 1: Identify the relevant electricity systems;

This project site is in Hebei Province of China, which belongs to North China Power Grid (NCPG) according to the public delineation of DNA, so NCPG is identified as the relevant electric system.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional);

For this project, Option I (only grid power plants are included in the calculation) is chosen.

Step 3: Select a method to determine the operating margin (OM);

Calculation of Operating Margin should be based on one of the four following methods according to the tool:

(a) Simple OM, or

(b) Simple adjusted OM, or

(c) Dispatch Data Analysis OM, or

(d) Average OM.

As the low-cost / must-run resources constituted less than 50% of total electricity generation of NCPG in recent five years (respectively 10.10%, 8.94%, 7,21%,6,39% and 5,92% in 2017, 2016, 2015, 2014 and 2013), the Simple OM (a) method is selected and the following data vintage is chosen to calculate the emission factor:

Ex ante option: use a 3-year generation-weighted average, based on the most recent data available, without requirement to monitor and recalculate the emissions factor during the crediting period. And according to the tool, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the emissions factor during the crediting period is required.

Step 4: Calculate the operating margin emission factor according to the selected method;

The Simple OM emission factor ($EF_{OM,y}$) is calculated as the generation-weighted average emissions per electricity unit (tCO_2e/MWh) of all generating sources serving in the system, excluding low-operating cost and must-run power plants. It may be calculated:

Option A: Based on the net electricity generation and a CO_2 emission factor of each power plant / unit, or

Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Option B can only be used if:

(a) The necessary data for Option A is not available; and

(b) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and

(c) Off-grid power plants are not included in the calculation.

In this project, all of the above conditions can be met, so Option B was chosen.

Under this option, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low-cost/must-run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMsimple,y} = \frac{\sum_i (FC_{i,y} \cdot NCV_{i,y} \cdot EF_{CO_2,i,y})}{EG_y}$$

(5)

Where:

$EF_{grid,OMsimple,y}$ = Simple operating margin CO_2 emission factor

		in year y (t CO_2/MWh)
$FC_{i,y}$	=	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO_2 emission factor of fuel type i in year y (t CO_2/GJ)
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in **Step 3.**

Based on the most recent three years (2015-2017) where the data are the latest and available at the time of this PD submission, the calculation result of $EF_{grid,OM,,y}$ is 0.9419 tCO_2e/MWh. The data is published by Ministry of Ecology and Environment of the People's Republic of China[2].

Step 5: Calculate the build margin (BM) emission factor;

As per Section 6.5 of TOOL07 (version 07.0), in terms of vintage of data, project participants can choose between one of the following two options:

(a) Option 1 - for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of CDM-PDD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period;

(b) Option 2 - For the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

In line with China's Baseline emission factors of regional grids 2019 (BEF_{2019}) published by the Development and Reform Commission of China, Option 1 is chosen for the project; the BM emission factor is calculated ex ante based on the most recent information available on units already built for sample group m at the time of this project description submission.

The sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

(a) Identify the set of five power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently ($SET_{5\text{-}units}$) and determine their annual electricity generation ($AEG_{SET\text{-}5\text{-}units}$, in MWh);

(b) Determine the annual electricity generation of the proposed project electricity system, excluding power units registered as CDM project activities (AEG_{total}, in MWh). Identify the set of power units, excluding power units registered as CDM project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) (SET≥20%) and determine their annual electricity generation ($AEG_{SET\geq 20\%}$, in MWh);

(c) From $SET_{5\text{-}units}$ and SET≥20% select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use SET_{sample} to calculate the build margin. In this case ignore Steps (d), (e) and (f).

Otherwise:

(d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as CDM project activities, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set ($SET_{sample\text{-}CDM}$) the annual electricity generation ($AEG_{SET\text{-}sample\text{-}CDM}$, in MWh); If the annual electricity generation of that set comprises at least 20% of the annual electricity generation of the proposed project electricity system (i.e. $AEG_{SET\text{-}sample\text{-}CDM} \geq 0.2 \times AEG_{total}$), then use the sample group $SET_{sample\text{-}CDM}$ to calculate the build margin. Ignore steps (e) and (f).

Otherwise:

(e) Include in the sample group $SET_{sample-CDM}$ the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the proposed project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sample-CDM->10yrs}$).

The BM emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which electricity generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m (EG_{m,y} \times EF_{EL,m,y})}{\sum_m EG_{m,y}} \quad (6)$$

where:

$EF_{grid,BM,y}$	=	Build margin emission factor of NCPG (tCO_2/MWh)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y ($t\ CO_2/MWh$)
m	=	Power units included in the build margin
y	=	The most recent year for which the generation data is available

As it is difficult to obtain the detailed data on the power generation, fuel consumption and thermal efficiency of each newly built power unit from public documents, a deviation of TOOL07 (07.0) is adopted following the clarifications given by the CDM EB concerning the BM emission factor calculation:

(1) The CDM EB suggested using the efficiency level of the best technology commercially available in the provincial/regional or national grid of China, as a conservative proxy, for each fuel type in estimating the fuel consumption to estimate the build margin.

(2) The EB agreed the use of capacity additions during last 1 ~ 3 years for estimating the build margin emission factor for grid electricity.

(3) The EB also agreed to use of weights estimated using installed capacity in place of annual electricity generation.

The newly built power plants in the past few years are bundled into “grouped new power plant” according to their construction year, their province and their fuel type. The annual net electricity generation in the year y of each "grouped new power plant" $EG_{m,y}$ is estimated according to their total capacity and the average utilization hours, as the following equation:

$$EG_{m,y} = CAP_m \times H_{m,y} \tag{7}$$

where:

$EG_{m,y}$	=	Annual net electricity generation the unit m in year y (MWh)
CAP_m	=	Installed capacity of the unit m (MW)
$H_{m,y}$	=	Utilization hour of the unit m in the year y (h), determined according to the average utilization hour of the same type of unit in the same province
y	=	The most recent year for which the generation data is available. For the calculation of BM in 2019, y = 2017
m	=	grouped new power plant

Since the newly built power plants in the same province (A), in the same year (t) and using the same fuel type (k) are grouped into "a grouped new power plant", CAP_m represents the total installed capacity of fuel type k power plants located in the provinces A and in the year t:

$$CAP_m = CAP_{A,t,k} \tag{8}$$

where:

CAP_m	=	Installed capacity of the unit m (MW), with m representing the specified combination of A, t, and k
$CAP_{A,t,k}$	=	Total installed capacity of fuel type k power plants located in the province A and in the year t
A	=	Provinces covered by the NCPG, namely, Beijing City, Tianjin City, Hebei Province, Shanxi Province, Shandong Province and Inner Mongolia Autonomous Region
t	=	Years related to the grouped new power plants, for the 2019 calculation, t represents 2017, 2016, 2015…. Until the aggregated electricity generation of the grouped new power plants reaches 20% of the total electricity generation of the NCPG

k	=	Fuel type of the grouped new power plants, including hydro, thermal (coal, gas, oil, waste incineration, other thermal), nuclear, wind, solar and others.

Figure 4 shows the procedure to determine the sample group of power units m.

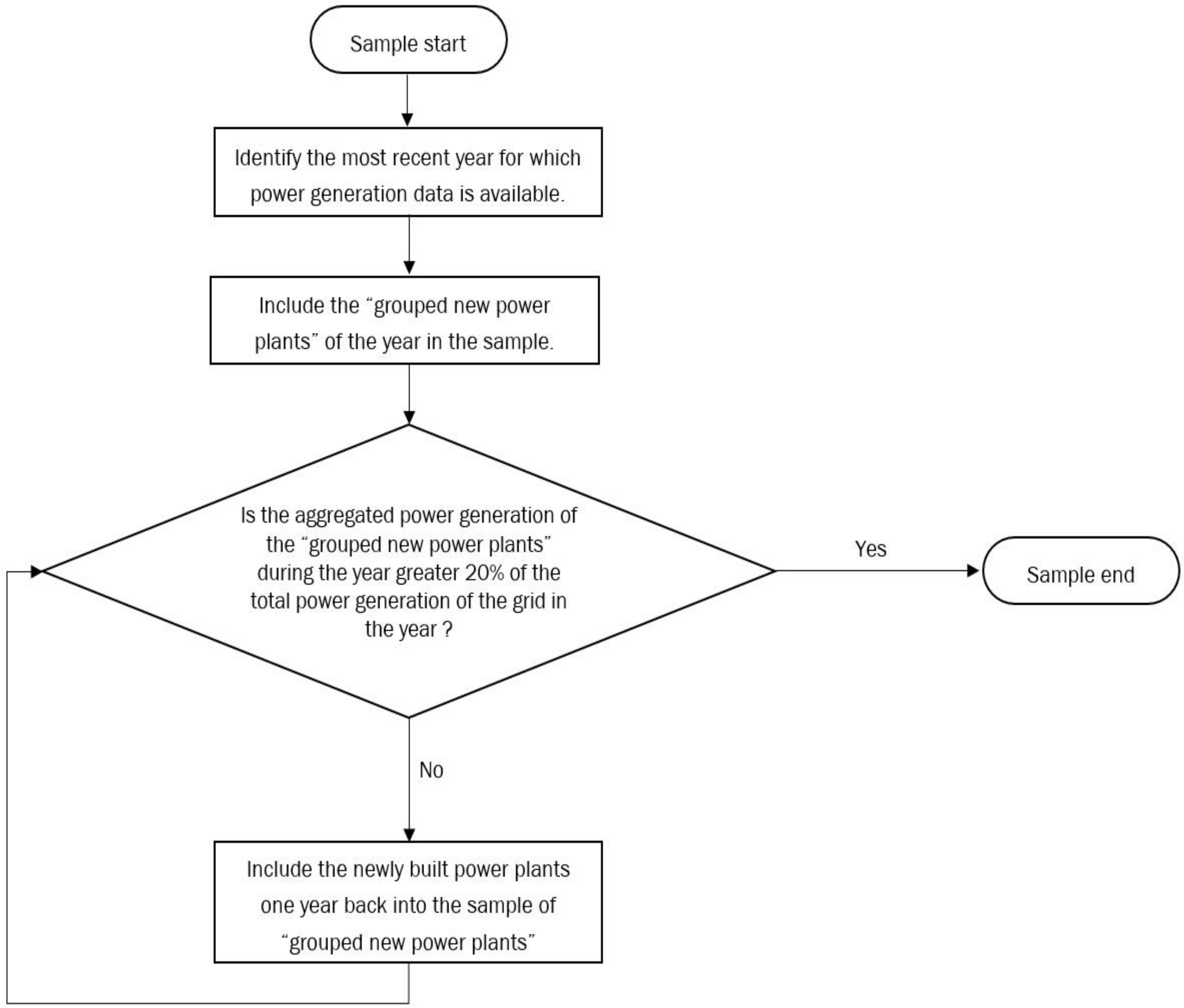

Figure 4 Procedure to determine the sample group of power units m used for the BM emission factor calculation

The emission factors of each fuel type $EF_{EL,m,y}$ are determined according to the Option A2 in the TOOL07, as the following equation:

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}} \quad (9)$$

where:

$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y (t CO_2/MWh)
$EF_{CO2,m,i,y}$	=	Average CO_2 emission factor of fuel type i used in power unit m in year y (t CO_2/GJ)
$\eta_{m,y}$	=	Average net energy conversion efficiency of power unit m in year y (ratio)
m	=	All power units serving the grid in year y except low-cost / must-run power units
3.6	=	Conversion factor (GJ/MWh)

Among the fuel types, the emission factors of hydro, nuclear, wind, solar, other thermal and others are 0. Concerning the emission factors of coal, gas, oil and waste incineration, Equation takes the following form due to conservativeness:

$$EF_{best,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{best,y}} \quad (10)$$

where:

$EF_{best,m,y}$	=	Emission factor of power unit m with the best technology commercially available in year y (t CO_2/MWh)
$\eta_{best,y}$	=	Power generation efficiency of the best technology commercially available in year y
m	=	Power units serving the grid with coal, gas, oil or waste incineration in year y

According to the latest and available data at the time of this PD submission, $EF_{grid,BM,y}$ is calculated to be 0.4819 tCO_2e/MWh. The data is published by Ministry of Ecology and Environment of the People's Republic of China.

Step 6: Calculate the combined margin (CM) emission factor.

The calculation of the combined margin emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

(a) Weighted average CM; or

(b) Simplified CM.

The weighted average CM method (option A) should be used as the preferred option.

The simplified CM method (option b) can only be used if:

a) The project activity is located in: (i) a Least Developed Country (LDC); or in (ii) a country with less than 10 registered CDM projects at the starting date of validation; or (iii) a Small Island Developing States (SIDS); and

b) The data requirements for the application of step 5 above cannot be met.

This Project choose option A.

The combined margin emissions factor is calculated as follows:

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM} \quad (11)$$

Where:

$EF_{grid,OM,y}$	=	a)	operating margin emission factor of NCPG (tCO_2e/MWh)
$EF_{grid,BM,y}$	=	b)	build margin CO_2 emission factor of NCPG (tCO_2e/MWh)
w_{OM}	=	c)	the weighting of operating margin emission factor (%)
w_{BM}	=	d)	the weighting of build margin emission factor (%)

According to the tool, for all other projects (except wind and solar power generation project activities): w_{OM} = 0.5 and w_{BM} = 0.5 for the first crediting period, and w_{OM} = 0.25 and w_{BM} = 0.75 for the second and third crediting period, unless otherwise specified in the approved methodology which refers to this tool.

As a wastes recycling project, w_{OM} = 0.5 and w_{BM} = 0.5 for the first crediting period.

$EF_{grid,CM,,y}$ = 0.9419 * 0.5 + 0.4819 * 0.5 = 0.7119 tCO_2e/MWh.

Base on the above parameters and equations, via checking the Joint PD-MR/3/ and ER calculation sheet/4/, CCSC confirmed that the baseline emissions (BE_y) is calculated as below:

Table3-1: Ex-ante value of parameters to calculate BE_y

Parameter	Value	Data sources
GWP_{CH4}	28 tCO_{2e}/tCH_4	IPCC Fifth Assessment Report (AR5)
D_{CH4}	0.67 kg/m^3	AMS-III.D
UF_b	0.94	AMS-III.D
MCF_j	76%	Table 10.17 of 2019 refinement to 2006

		IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
$B_{o,LT}$	Dairy cattle:0.13 m^3CH_4/kg-VS	Table 10.16 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
$VS_{default}$	Dairy cattle: 3.474 kg/hd/day	$VS_{default}$ =$VS_{rate(T)}$ * TAM / 1000 = 9.0 * 386 / 1000 TAM:Default Values for livestock weight for Animal Categories in Asia area is 386kg(mean) for dairy cattle $VS_{rate(T)}$ = Default Values for Volatile Solid Excretion Rate, 9.0 kg VS 1000 kg animal mass^{-1} day^{-1} for dairy cattle sourced from Volume 4, chapter 10, table 10.13A of 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories.
$MS\%_{Bl,j}$	100%	FSR
nd_y	365 days	FSR
$N_{da,y}$	365 days	FSR
$N_{p,y}$	20,000	FSR

The result of the ex-ante calculated baseline emissions of methane from the manure treatment processes of the project is:

Table 3-2: Ex-ante Baseline Emissions BE_y

Index for all types of livestock (LT)	The annual average number of animals ($N_{LT,y}$)	Volatile solids production of livestock LT in year y ($VS_{LT,y}$) (kg-dm/animal/year)	Baseline Emissions (BE_y) tCO_2e
-	$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365})$	$VS_{LT,y} = VS_{default} \times nd_y$	$BE_y = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum_{j,LT} MCF_j \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{Bl,j}$
Dairy cattle	20,000	1268.01	44,184
Total	-	-	44,184

As per the FSR, the annual exported electricity to the NCPG of the project is expected to be 27,000 MWh, the combined margin CO_2 emission factor for the grid is 0.7119 tCO_2/MWh, hence the expected annual baseline emissions associated with electricity generation of the project is calculated as follows:

$BE_{elec,y} = EG_{PJ,y} \times EF_{grid,y}$ = 27,000 MWh× 0.7119 tCO_2/MWh = 19,221 tCO_2

$BE_y = BE_{CH4,y} + BE_{elec,y}$ = 44,184 tCO_2e + 19,221 tCO_2e = 63,405tCO_2e

3.4.6.2 Project emissions

Via checking the applied methodology (AMS-III.D ver. 21.0 and AMS-I.D ver. 18.0) , project emissions consist of:

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y} \tag{12}$$

Where:

PE_y	=	Project emissions in year *y* (t CO_2e)
$PE_{PL,y}$	=	Emissions due to physical leakage of biogas in year *y* (t CO_2e)
$PE_{flare,y}$	=	Emissions from flaring or combustion of the biogas stream in the year *y* (t CO_2e)
$PE_{power,y}$	=	Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year *y* (t CO_2e)
$PE_{transp,y}$	=	Emissions from incremental transportation in the year *y* (t CO_2e), as per relevant paragraph in AMS-III.AO
$PE_{storage,y}$	=	Emissions from the storage of manure (t CO_2e)

1. Emissions due to physical leakage of biogas in year y

Project emissions due to physical leakage of biogas from the animal manure management systems used to produce, collect and transport the biogas to the point of flaring or gainful use are estimated as:

$$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{i,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y} \tag{13}$$

Where:

GWP_{CH4}	=	Global Warming Potential (GWP) of CH_4 applicable to the crediting period (t CO_2e/t CH_4)
D_{CH4}	=	CH_4 density (0.00067 t/m^3 at room temperature (20 °C) and 1 atm pressure)
LT	=	Index for all types of livestock

i = Index for animal manure management system

$N_{LT,y}$ = Annual average number of animals of type LT in year y (numbers)

$VS_{LT,y}$ = Volatile solids production/excretion per animal of livestock LT in year y (on a dry matter weight basis, kg-dm/animal/year)

$MS\%_{i,y}$ = Fraction of manure handled in system i in year y. If the project activity involves sequential manure management systems, the procedure specified in paragraph 18(e) of AMS-III.D shall be used to estimate the project emissions due to physical leakage of biogas in each stage

$B_{0,LT}$ = Maximum methane producing potential of the volatile solid generated for animal type LT (m^3 CH_4/kg-VS), as per paragraph18 of AMS-III.D

2. **Emissions from flaring or combustion of the biogas stream in the year y**

According to methodology, in the case of flaring of the recovered biogas, project emissions are estimated using the procedures described in the methodological tool "Project emissions from flaring". If the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. However, this use of the recovered biogas shall be included in the project boundary and its output shall be monitored in order to ensure that the recovered biogas is actually destroyed, even if the emission reductions from this component are not claimed.

For the project, only when emergency occurs, the emergency flare system will be used and the biogas will be flared. The project emission is calculated using the procedures described in the methodological TOOL06 "Project emissions from flaring" (version 04.0) as below:

$$PE_{flare,\ y} = GWP_{CH4} \times \sum_{m=1}^{525600} F_{CH4,RG,m} \times (1 - \eta_{flare,\ m}) \times 10^{-3} \qquad (14)$$

Where:

$PE_{flare,\ y}$ = Project emissions from flaring of the residual gas in year y (tCO_2e)

GWP_{CH4} = Global warming potential of methane valid for the commitment period (tCO_2e/tCH_4)

$F_{CH4,\ RG,\ m}$ = Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{flare,\ m}$ = Flare efficiency in the minute m

According to **Option A** in the "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (version 03.0), the calculation equation of mass flow of residual gas is listed directly as follows:

$$F_{CH4,RG,m} = V_{RG,t,db} \times \nu_{RG,t,db} \times \rho_{RG,n} \times \frac{h}{1000} \quad (15)$$

With:

$$\rho_{RG,n} = \frac{P_n \times MM_i}{R_u \times T_n} \quad (16)$$

Where:

$F_{CH4,RG,m}$	=	Mass flow of greenhouse gas I in the in the residual gaseous stream in time interval t (kg gas/m)
$V_{RG,t,db}$	=	Volumetric flow of the gaseous stream in time interval t on a dry basis (m³ dry gas/h)
$\nu_{RG,t,db}$	=	Volumetric fraction of greenhouse gas i in the gaseous stream in a time interval t on a dry basis (m³ gas i/m³ dry gas) , i =Residual gas (RG), $\nu_{RG,t,db}$=$\nu_{i,t,db}$
$\rho_{RG,n}$	=	Density of greenhouse gas i in the gaseous stream in time interval t (kg gas i/m³ gas i), i =Residual gas (RG)
P_n	=	Absolute pressure of the gaseous stream in time interval t (Pa)
MM_i	=	Molecular mass of greenhouse gas i (kg/kmol), i =Residual gas (RG)
R_u	=	Universal ideal gases constant (Pa.m^3 /kmol.K)
T_n	=	Temperature of the gaseous stream in time interval t (K)
h	=	The total operation hours of the flare (hours)

Since this is a small scale project, a default value for the fraction of methane in the biogas ($f_{CH4,default}$) is applied to substitute the parameter $\nu_{RG,t,db}$. And $F_{CH4,\ RG,\ m}$ will be monitored ex post in the verification period.

Based on the FSR, all the methane generated by the project will be used as energy supply. In normal circumstances, the project proponent would like to consider a methane destruction efficiency of 100% of the generation system since all the biogas will be utilized and there will be no excess biogas flared. In order to ensure no biogas is released under exigencies, an enclosed emergency flare system is installed at the project site, this emergency flare system is not used under normal operation.

In the case of enclosed flares, project participants may choose between the following two options to determine the flare efficiency for minute m ($\eta_{flare,m}$) and shall document in the VCS-PD which option is selected: (a) Option A: Apply a default value for flare efficiency; (b) Option B: Measure the flare efficiency. The project applied Option A to determine the value of $\eta_{flare,m}$.

In case this emergency flare system operated, the $PE_{flare,y}$ will be monitored and calculated ex post. The estimated $PE_{flare,y}$ is 0 tCO_2e ex ante.

3. Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y

Project emissions from electricity and fossil fuel consumption are determined by following the methodological tool "Project and leakage emissions from anaerobic digesters" (version 02.0), where $PE_{Power,y}$ is the sum of $PE_{EC,y}$ and $PE_{FC,y}$ in the tool.

The project does not use any fossil fuel, so $PE_{FC,y}$ is not included in the project. Thus, $PE_{FC,y}$ = 0 tCO_2e.

According to methodological tool "Project and leakage emissions from anaerobic digesters", $PE_{EC,y}$ shall be calculated using the "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0) as follows:

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EL,j,y} \times (1 + TDL_{j,y}) \quad (17)$$

Where:

$PE_{EC,y}$	=	Project emissions from electricity consumption in year y (t CO_2 / yr)
$EC_{PJ,j,y}$	=	Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)
$EF_{EL,j,y}$	=	Emission factor for electricity generation for source j in year y (t CO_2/MWh)
$TDL_{j,y}$	=	Average technical transmission and distribution losses for providing electricity to source j in year y
j	=	Sources of electricity consumption in the project

As stated above, according to the tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation"(version 03.0), for the project is in the case of Scenario A: Electricity consumption from the grid, the project participants choose Option A1: Calculate the combined margin emission factor of the applicable electricity system, using the procedures in the latest approved version of the "Tool to calculate the emission factor for an electricity system" (version 07.0) ($EF_{EL,j,y} = EF_{grid,CM,y}$).

According to the tool "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" (version 03.0), the project belongs to the case of Scenario A, use as default values of 20% for project emission, i.e. $TDL_{j,y}$ =20%.

$EC_{PJ,j,y}$ is ex ante determined as 0 in the PD and will be monitored ex post in the verification period.

4. Emissions from incremental transportation in the year y

According to AMS-III.D, emission from transportation of animal manure between livestock farms and the project site is calculated using relevant paragraph in AMS-III.AO as below:

$$PE_{transp,y} = (Q_y/CT_y)*DAF_w*EF_{CO2,transport} + (Q_{res\,.\,waste,y}/CT_{res\,.\,waste,y})*DAF_{res\,.\,waste}*EF_{CO2,transport} \quad (18)$$

Where:

Q_y	=	Quantity of raw waste/manure treated and/or wastewater co-digested in the year y(tonnes)
CT_y	=	Average truck capacity for transportation (tonnes/truck)
DAF_w	=	Average incremental distance for raw solid waste/manure and/or wastewater transportation (km/truck)
$EF_{CO2,\ transport}$	=	CO_2 emission factor from fuel use due to transportation ($kgCO_2$/km, IPCC default values or local values may be used)
$Q_{res\,.\,waste,y}$	=	Quantity of residual waste produced in year y (tonnes)
$CT_{res,waste,y}$	=	Average truck capacity for residual waste transportation (tonnes/truck)
$DAF_{res\,.\,waste}$	=	Average distance for residual waste transportation (km/truck)

The project site is at the east side of the pasture and waste from the pasture will all enter the receiving tank of the pre-treatment system through tight pipeline, thus there is no emission from transportation of animal manure between livestock farms and the project site. $PE_{transp,y} = 0$.

5. Emissions from the storage of manure

Project emissions on account of storage of manure before being fed into the anaerobic digester shall be accounted for if both condition (a) and condition (b) below are satisfied:

(a) The storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester;

(b) The dry matter content of the manure when removed from the animal barns is less than 20%.

1. Through satellite maps and on-site visits, it can be confirmed that the breeding farm is very close to the ranch, right next to it.

2. By checking the FSR and EIA of the project, the storage time of the manure after removal from the animal barns, including transportation, is within 24 hours before being fed into the anaerobic digester, and there are no related emissions during the entire project's raw material collection and other processes. Thus, there will be no $PE_{storage}$ related emissions.

3.The on-site audit has visited the relevant personnel and confirmed that the manure was transported into the system through a closed transportation pipeline within 5~7 hours, not exceed 24 hours. Therefore, condition(a) are not satisfied and project emissions from the storage of manure is not accounted for, $PE_{storage,y}$ = 0 tCO_2e.

Base on the above parameters and equations, via check the Joint PD-MR and ER calculation sheet, CCSC confirmed that the Project emissions (PE_y) is calculated as:

Table 3-3: Ex-ante value of parameters to calculate project emissions

Parameter	Value	Data sources
GWP_{CH4}	28 tCO_2e/tCH_4	IPCC Fifth Assessment Report (AR5)
D_{CH4} (ρ_{CH4})	0.67 kg/m^3	AMS-III.D
$B_{o,LT}$	Dairy cattle: 0.13 m^3CH_4/kg-VS	Table 10.16 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
$VS_{default}$	Dairy cattle: 3.474 kg/hd/day	$VS_{default}$ =$VS_{rate(T)}$ * TAM / 1000 = 9.0 * 386 / 1000 TAM: Default Values for livestock weight for Animal Categories in Asia area is 386kg(mean) for dairy cattle $VS_{rate(T)}$ = Default Values for Volatile Solid Excretion Rate, 9.0 kg VS 1000 kg animal $mass^{-1}$ day^{-1} for dairy cattle sourced from Volume 4, chapter 10, table 10.13A of 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories.
$MS\%_{i,y}$	100%	FSR
$N_{da,y}$	365 days	FSR
$N_{p,y}$	20,000	FSR
$f_{CH4,\ default}$	0.6	"Tool to Project and leakage emissions from anaerobic digesters"

Emissions due to physical leakage of biogas in year y

As described above, project emissions from physical leakage of biogas are calculated as per equations (12), (13).

Table 3-4: Ex-ante calculate Physical leakage of biogas ($PE_{PL,y}$)

Index for all types of livestock (LT)	The annual average number of animals ($N_{LT,y}$)	Volatile solids production of livestock LT in year y ($VS_{LT,y}$) (kg-dm/animal/year)	Physical leakage of biogas ($PE_{PL,y}$) tCO_2e
-	$N_{LT,y} = N_{da,y} \times (\frac{N_{p,y}}{365})$	$VS_{LT,y} = VS_{default} \times nd_y$	$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{j,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y}$
Dairy cattle	20,000	1,268.01	6,185
Total	-	-	6,185

So, the ex-ante calculated result of the project's Physical leakage of biogas is 6,185 tCO_2e.

Emissions from flaring or combustion of the biogas stream in the year y

As described above, in case this emergency flare system operated, the $PE_{flare,y}$ will be monitored and calculated ex post. The estimated $PE_{flare,y}$ is 0 tCO_2e ex ante.

Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y

As described above, the project does not use any fossil fuel, so $PE_{FC,y}$ is not included in the project emission.

As described above, the electricity consumption of the project site is ex ante determined as 0 in the PD and will be monitored ex post in the verification period.Thus, $PE_{EC,y}$ = 0.

Emissions from incremental transportation in the year y

As described above, $PE_{transp,y}$ is excluded, hence $PE_{transp,y}$ = 0 for the project.

Emissions from the storage of manure

As described above, $PE_{storage,y}$ is excluded, hence emissions from the storage of manure are not accounted for, $PE_{storage,y}$ = 0 tCO_2e.

So, the ex-ante estimated project emission shall calculate as per equation (12):

$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y} = PE_{PL,y} = 6,185 tCO_2e$

3.4.6.3 Leakage

As per "Project and leakage emissions from anaerobic digesters" (version 02.0), the leakage emissions associated with the anaerobic digester depend on how the digestate is managed. The leakage emissions include emissions associated with storage of digestate and composting of the digestate.

Via site inspection and checking the FSR/5/, CCSC confirmed that Digestate from the anaerobic digesters of the project will be used to produce organic fertilizer as soon as digestate is removed from the digesters. Digestate will not be stored under anaerobic conditions. In addition, the project does not involve composting of digestate. Therefore, LE_y=0 tCO_2

3.4.6.4 Emission Reductions

According to ASM III-D Version 21.0 and ASM I-D Version 18.0, emission reductions are calculated as follows:

$$\mathrm{ER_y} = \mathrm{BE_y} - \mathrm{PE_y}$$

Where:

ER_y = Emission reductions in year y (tCO_2e/yr)

BE_y = Baseline emissions in year y (tCO_2e/yr)

PE_y = Project emissions in year y (tCO_2e/yr)

Hence for the project activity, the estimated amount of GHG emission reductions (ERy) is 400,540 tCO_2e during the crediting period from 21-March-2021 to 20-March-2028, resulting in estimated average annual emission reductions of 57,220 tCO_2e.

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
21-Mar-2021~ 31-Dec-2021	49,682	4,846	0	44,835
1-Jan-2022~ 31-Dec-2022	63,405	6,185	0	57,220
1-Jan-2023~ 31-Dec-2023	63,405	6,185	0	57,220
1-Jan-2024~ 31-Dec-2024	63,405	6,185	0	57,220
1-Jan-2025~ 31-Dec-2025	63,405	6,185	0	57,220
1-Jan-2026~ 31-Dec-2026	63,405	6,185	0	57,220
1-Jan-2027~	63,405	6,185	0	57,220

31-Dec-2027				
1-Jan-2028~ 20-Mar-2028	13,723	1,339	0	12,385
Total	443,835	43,295	0	400,540

neral, the assessment team confirms the following:

- All assumptions and data used by the project participants are listed in the Joint PD-MR (version 02.0 dated 06-Mar-2023) and/or supporting documents, including their references and sources;
- All documentation used by the project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the Joint PD-MR (version 02.0 dated 06-Mar-2023);
- All values used in Joint PD-MR (version 02.0 dated 06-Mar-2023) are considered reasonable in the context of the project activity;
- The baseline methodology has been applied correctly to calculate project emissions, baseline emissions, and leakage emissions;
- All estimates of the baseline, project and leakage emissions can be replicated using the data and parameter values provided in Joint PD-MR (version 02.0 dated 06-Mar-2023).

CCSC confirms that the methodologies and the above referenced tools have been applied correctly to calculate baseline emissions, project emissions, leakage and net GHG emission reductions and removals.

3.4.7 Methodology Deviations

As plant specific fuel consumption and electricity generation data are not publicly available in China, the Guidance caused by DNV's request for deviation of Chinese project activities for the CDM baseline methodology AM0005 has been applied for calculating the build margin (BM) emission factor of this project activity. The deviation has no impact on conservativeness of the quantification of GHG emission reductions or removals.

3.4.8 Monitoring Plan

The approved baseline and monitoring methodology AMS-III.D (ver. 21.0) and AMS-I.D (ver. 18.0), have been applied. The monitoring diagram of the project is shown in Figure 3-1.

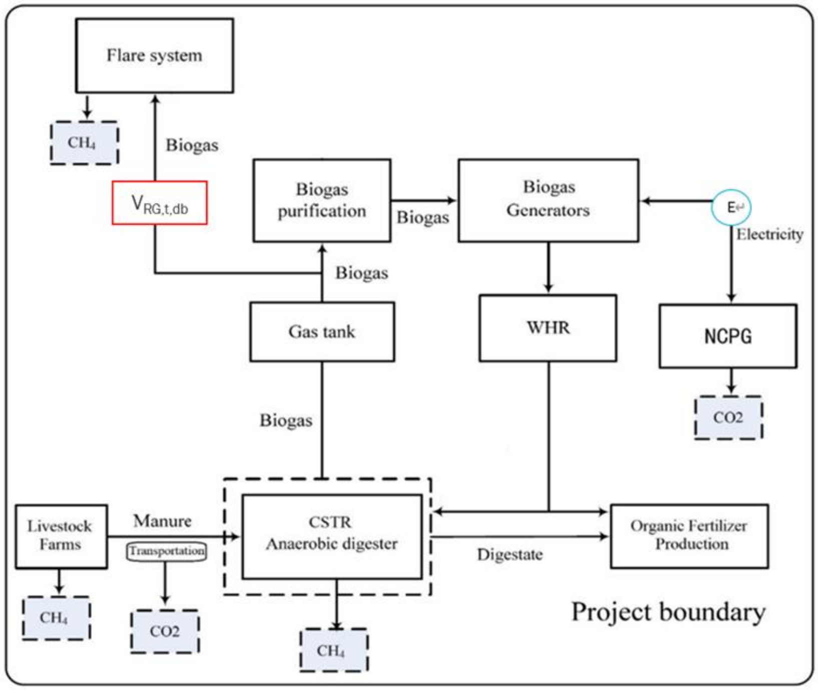

Figure 3-1 Project monitoring diagram

The monitoring plan is in accordance with the monitoring methodology; the monitoring plan will give opportunity for real measurement of achieved emission reductions. CCSC has checked all the parameters presented in the monitoring plan against the requirements of the methodology; no deviations relevant to the project have been found in the plan.

CCSC confirms that the monitoring arrangements described in the monitoring plan are feasible within the project design, and the means of implementation of the monitoring plan are sufficient to ensure the emission reductions achieved by/resulting from the project can be reported ex post and verified.

1. Data and parameters available at validation

Please refer to the following tables for assessment of each parameter determined ex-ante only used to calculate the ex-ante emission reduction:

Data / Parameter	GWP_{CH4}
Data unit	tCO_2e/tCH_4
Description	Global Warming Potential (GWP) of CH_4 applicable to the crediting period
Source of data	IPCC

Value applied	28

Data / Parameter	D_{CH4}
Data unit	kg/m^3
Description	CH_4 density
Source of data	AMS-III.D
Value applied	0.67 (at 20 °C and 1 atm pressure)

Data / Parameter	UF_b
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.D
Value applied	0.94

Data / Parameter	MCF_j
Data unit	-
Description	Annual methane conversion factor (MCF) for the baseline animal manure management system j
Source of data	2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	76% Since no country or regional specific value is available. Default value from table 10.17 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 is applied. By checking public information, it is confirmed that the baseline site where anaerobic manure treatment facility is located in warm temperate moist region (with annual average temperature around 13°C and ratio of potential evapotranspiration to precipitation is less than 1), as per IPCC default values provided in Table 10.17 of 2019 Refinement to the 2006 IPCC Guidelines Volume 4 Chapter 10, a value of 76% for warm temperate moist region is applied.

Data / Parameter	$B_{0,LT}$
Data unit	m^3 CH_4/kg-VS
Description	Maximum methane producing potential of the volatile solid generated for animal type LT
Source of data	Table 10.16 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
Value applied	<table><tr><td>Animal type</td><td>$B_{0,LT}$</td></tr><tr><td>Dairy cattle</td><td>0.13</td></tr></table> Since no country or regional specific value is available. Default values from table 10.16 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10 Chapter 10 is applied. The species of the dairy cow in this project is Chinese Holstein which belongs to 'Asia' category. Value for $B_{0,\ LT,y}$ is 0.13 m^3CH_4/kg-VS.

Data / Parameter	$MS\%_{BL,i}$
Data unit	-
Description	Fraction of manure handled in baseline animal manure management system j
Source of data	FSR
Value applied	100%

Data / Parameter	$N_{da,y}$
Data unit	Day
Description	Number of days animal is alive in the farm in the year y
Source of data	Project proponents
Value applied	365

Data / Parameter	$VS_{default}$

Data unit	kg/hd/day
Description	Default value for the volatile solid excretion rate per day on a dry-matter basis for a defined livestock population
Source of data	2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	<table><tr><td>Animal type</td><td>$VS_{default}$</td></tr><tr><td>Dairy cattle</td><td>3.474</td></tr></table> Calculated via equation 10.22A in 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories. $VS_{default} = VS_{rate(T)} * TAM / 1000 = 9.0 * 386 / 1000$ TAM: Default Values for livestock weight for Animal Categories in Asia area is 386kg(mean) for dairy cattle $VS_{rate(T)}$ = Default Values for Volatile Solid Excretion Rate, 9.0 kg VS 1000 kg animal mass^{-1} day^{-1} for dairy cattle sourced from Volume 4, chapter 10, table 10.13A of 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories.

Data / Parameter	$EF_{grid,OM,y}$
Data unit	t CO_2/MWh
Description	Operating margin CO_2 emission factor in year y
Source of data	"2019 Baseline Emission Factors for Regional Power Grids in China" published by Ministry of Ecology and Environment of China
Value applied	0.9419

Data / Parameter	$EF_{grid,BM,y}$
Data unit	t CO_2/MWh
Description	Build margin CO_2 emission factor in year y
Source of data	"2019 Baseline Emission Factors for Regional Power Grids in China" published by Ministry of Ecology and Environment of

	China
Value applied	0.4819

Data / Parameter	$EF_{grid,CM,y}$ ($EF_{grid,y}$, $EF_{EL,j,y}$)
Data unit	t CO_2/MWh
Description	Combined margin emission factor for the grid in year y
Source of data	Calculated based on "Tool to calculate the emission factor for an electricity system" (version 07.0).
Value applied	0.7119

Data / Parameter	NCV_{CH4}
Data unit	MJ/Nm^3
Description	NCV of methane
Source of data	AMS-III.D
Value applied	35.9

Data / Parameter	$TDL_{j,y}$
Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to source j in year y
Source of data	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)
Value applied	20%

Data / Parameter	EE_y
Data unit	%

Description	Energy Conversion Efficiency of the project equipment
Source of data	AMS-III.D
Value applied	40%

Data / Parameter	$SPEC_{flare}$
Data unit	Temperature - °C Flow rate or heat flux - kg/h or m^3 /h Maintenance schedule - number of days
Description	Manufacturer's flare operating specifications for temperature, flow rate and maintenance schedule
Source of data	Flare manufacturer
Value applied	Temperature – 500~1270°C Flow rate or heat flux - 1100 m^3 /h Maintenance schedule - every 12-24 months (around 365days – 730 days) based on operating conditions

Data / Parameter	$\eta_{flare,m}$
Data unit	%
Description	Flare efficiency in minute m
Source of data	Default value of methodological tool 06" Project emissions from flaring (Version 04.0)"
Value applied	When the flame is detected in the minute m and the temperature of the flare and the flow rate of the residual gas to the flare ($F_{RG,m}$) is within the manufacturer's operating specification for the flare ($SPEC_{flare}$) in the minute m: 90% Otherwise: 0%

Data / Parameter	$f_{CH4,default}$
Data unit	m^3 CH_4 / m^3 biogas corrected to reference conditions

Description	Default value for the fraction of methane in the biogas
Source of data	The default value was derived based on reported values from registered projects and research papers (Davidsson, 2007)
Value applied	0.6

Data / Parameter	R_u
Data unit	$Pa.m^3/kmol.K$
Description	Universal ideal gases constant
Source of data	"Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)
Value applied	8,314

Data / Parameter	MM_i
Data unit	kg/kmol
Description	Molecular mass of greenhouse gas CH_4
Source of data	"Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)
Value applied	16.04

Data / Parameter	T_n
Data unit	K
Description	Temperature at normal conditions
Source of data	"Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)
Value applied	273.15

Data / Parameter	$\rho_{i,n}$ & $\rho_{RG,n}$
Data unit	Kg gas i/m^3 gas I, i=RG=CH_4
Description	Density of greenhouse gas CH_4 in the gaseous stream at normal conditions (0℃ and 1 atm)
Source of data	Calculated in the section 3.4.6 as per the latest version of "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)
Value applied	0.71566

2. Data and Parameters Monitored

It has been confirmed that the project site is at the east side of the pasture and waste from the pasture will all enter the receiving tank of the pre-treatment system through tight pipeline, thus there is no emission from transportation of animal manure between livestock farms and the project site. $PE_{transp,y}$ = 0. Thus, the related parameters of Q_y, CT_y, DAF_w, $Q_{res\,.\,waste,y}$, $CT_{res,waste,y}$, $DAF_{res\,.\,waste}$ are not been monitored.

The monitoring parameters required by the methodology and applicable tools for the project are summarized in the below table：

Data / Parameter	$MS\%_{i,y}$
Data unit	-
Description	Fraction of manure handled in system i in year y
Source of data	Project proponents
Description of measurement methods and procedures to be applied	all manure would be handled in this project, 100% is applied for $MS\%_{i,y}$ in ex-ante estimation. This parameter will be monitored in verification period.
Frequency of monitoring/recording	-
Value applied	100%
Monitoring equipment	-
QA/QC procedures to be applied	-

Purpose of data	Calculation of baseline emissions and project emissions
Calculation method	-

Data / Parameter	$N_{p,y}$
Data unit	Number
Description	Number of animals produced annually of type LT for the year y
Source of data	Project proponents
Description of measurement methods and procedures to be applied	The number of Dairy cattles produced in the farm will be recorded manually by the responsible staff.
Frequency of monitoring/recording	Annually, based on monthly records
Value applied	20,000 estimated ex ante and will be monitored ex post
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	To calculate the annual average number of animals ($N_{LT,y}$)
Calculation method	-

Data / Parameter	nd_y
Data unit	number
Description	The number of days the treatment plant was operational in year y.
Source of data	Project proponents
Description of measurement methods and procedures to be applied	365 days used for ex-ante estimation. The actual number of days the treatment plant was operationally used in the monitoring periods will be monitored and recorded by staff.
Frequency of monitoring/recording	Annually, based on daily records and monthly aggregation
Value applied	365

Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	To estimate the annual volatile solid excretions for livestock LT entering all animal waste management systems on a dry matter weight basis ($VS_{LT,y}$).
Calculation method	-

Data / Parameter	EG_y
Data unit	MWh
Description	Total electricity generated from the recovered biogas in year y
Source of data	Electricity meter
Description of measurement methods and procedures to be applied	The parameter is not monitored by the project owner, it will be replaced by the parameter $EG_{PJ,y}$.
Frequency of monitoring/recording	Continuously monitored and monthly recorded
Value applied	27,000 (Ex-ante estimation)
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	-
Purpose of data	Calculation of emission reductions
Calculation method	-

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh
Description	Quantity of electricity consumed by the project in year y
Source of data	Electricity meter
Description of measurement methods and procedures to be	Continuously measured by electricity meter and at least monthly recorded

applied	
Frequency of monitoring/recording	Continuously measured and at least monthly recorded
Value applied	0 (estimated ex-ante)
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	The data will be cross checked with Electricity Transaction Note. The metering instruments with accuracy no less than 0.5s will be calibrated annually in accordance with the latest version "Technical administrative code of electric energy metering".
Purpose of data	Calculation of project emissions
Calculation method	-

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh
Description	Quantity of electricity generation supplied by the project to the grid in year y
Source of data	Electricity meter
Description of measurement methods and procedures to be applied	Measured by electricity meter continuously and monthly recorded
Frequency of monitoring/recording	Continuous monitoring and at least monthly recording
Value applied	-
Monitoring equipment	Electricity meter
QA/QC procedures to be applied	The data will be cross checked with Electricity Transaction Note. The metering instruments with accuracy no less than 0.5s will be calibrated annually in accordance with the latest version "Technical administrative code of electric energy metering".
Purpose of data	Calculation of baseline emissions
Calculation method	-

Data / Parameter	$F_{CH4,RG,m}$
Data unit	t CH_4
Description	Mass flow of methane in the residual gas in the minute m
Source of data	Calculated
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	-
Value applied	0 when there is no malfunction and the flare system does not operate
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-

Data / Parameter	$V_{RG,t,db}$
Data unit	m^3 dry gas/h
Description	Volumetric flow of the residue gas in time interval t on a dry basis at normal conditions
Source of data	Measured by a flow meter continuously and converted automatically by the recovery monitoring system
Description of measurement methods and procedures to be applied	The parameters of $V_{t,db,n}$ is measured by a flow meter continuously, which will be converted automatically into standard value $V_{t,db,n}$ (at the normal condition of 0 ℃ and 101,325 Pa) by the recovery monitoring system
Frequency of monitoring/recording	Monitoring continuously and recording monthly
Value applied	The data will be monitored ex post in the monitoring period. In the estimated net GHG emission reduction phase, the amount of biogas is estimated and calculated as per section 3.4.6, please refer to ER calculation sheet for details.

Monitoring equipment	Flow meter
QA/QC procedures to be applied	Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory for all projects applying large scale methodology(ies). Calibration will be carried out once every two years according to manufacturer's specifications. <table><tr><th>Type</th><th>Serial number</th><th>Date of calibration</th><th>Validity</th></tr><tr><td>HGF-3000F</td><td>200051040051</td><td>06-Apr-2020;05-Apr-2022</td><td>05-Apr-2022;04-Apr-2024</td></tr></table>
Purpose of data	Calculation of project emissions
Calculation method	-

Data / Parameter	Operation hours (to calculate $F_{CH4,RG,m}$)
Data unit	Hours
Description	The total operation hours of the flare (to calculate $F_{CH4,RG,m}$)
Source of data	Project participants
Description of measurement methods and procedures to be applied	-
Frequency of monitoring/recording	Monitoring hourly
Value applied	The data will be monitored ex post in the monitoring period.
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of project emissions.
Calculation method	-

CCSC assessment team confirms that the monitoring plan contains all necessary parameters which have been clearly described in Joint PD-MR and that the means of monitoring described in the monitoring plan complies with the requirements of the methodology.

Data Management and Quality Control

A Management Structure of the Monitoring Team is provided in the Joint PD-MR/3/. The functions such as data collection, aggregation, verification, calculation, archiving, as well as the maintenance of equipment etc. have been defined. Quality assurance and quality control procedures for recording, maintaining and data archiving etc. will be ensured according to Verra rules. The monitoring system equipment will be implementation properly upon the agreement of the project company and the grid company. The monitoring data would be cross checked by the Electricity transaction note of the project for the purpose of quality control. The project owner will record the readings of the meter monthly. The calibration of the meters will be implemented as per relevant national standards. Data management and quality control system are quoted in the Joint PD-MR/3/.

In conclusion, based on document review, on-site inspection and stakeholder interview, together based on CCSC's local and sectoral expertise, CCSC confirms that:

The monitoring plan is in compliance with the requirements of the methodology.

Monitoring arrangements described in the monitoring plan are feasible within the project design.

The PP's ability to implement the monitoring plan can be guaranteed.

3.5 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

1. Assessment of Data and parameters available at validation

Via checking the Joint-PD-MR/37/, it is confirmed that all the ex-ante data and parameters are same to the Joint-PD-MR which is verified as correct. Refer to section 3.4.8 for the detail assessment of ex-ante parameters.

2. Assessment of Data and parameters monitored

During the verification all relevant monitoring parameters (as listed in chapter 3.4.8 of this report) have been verified with regard to the

(i) appropriateness of the applied measurement / determination method,

(ii) the correctness of the values applied for ER calculation,

(iii) the accuracy, and applied QA/QC measures.

Through the onsite visit, interviewing the project owner and reviewing the project operation log, CCSC confirmed that the project utilizes animal manure waste as ferment material to produce biogas in the Completely Stirred Tank Reactor (CSTR) type anaerobic digesters and the recovered biogas for electricity generation, which is located at Xiwang Village, Houguan Town, Weixian District, Xingtai City, Hebei Province, P.R China. During this monitoring period, the project provided total installed capacity of 2.33 MW which is consistent with the value in project description. The bidirectional electricity meter continuously measures the amount of electricity exported and imported. The monitoring team members record the data on a monthly basis. All the monitoring instrument was calibrated regularly for accuracy by qualified third-party organizations according to the national regulations during the monitoring period.

The project can reduce GHG emissions in two aspects: 1) it avoids methane emissions by capturing and destroying biogas which would have been directly vented into the atmosphere in the baseline scenario; 2) it reduces CO_2 emissions by displacing part of the electricity that would otherwise have been supplied by NCPG. In the monitoring period from 21-March-2021 to 30-September-2022, the total verified carbon units (VCUs) achieved in the monitoring period are 47,385 tCO_2e, which has been verified by on-site inspection, checking the related documentations and interviewing with the project owner.

Via checking the Joint PD-MR/1//3/ and by site inspection, CCSC verified that there are no changes on the key equipment and technology between project implementation and the project description.

This verification covers the period from 21-March-2021 to 30-Septemper-2022.

As part of the site visit, the assessment team was able to confirm that the project implementation is in accordance with the project description. There are no material discrepancies between project implementation and the project description in the Joint PD-MR V02/3/.

Via on-site interview with the project owner and checking the UNFCCC, Gold standard website, the assessment team confirms that the project has neither participated nor been rejected under any other GHG programs.

Via on-site interview with the project owner and checking REC website, the assessment team confirms that the project has not received or sought any other form of environmental credit, or become eligible to do so since validation or previous verification.

The net GHG emission reductions during this monitoring period from the project was not used for compliance with emission trading programs or to meet binding limits on GHG emissions confirmed via on-site interview with the project owner and checking the other GHG programs and other form of environmental credit website.

It is verified that the monitoring system is fully functional to generate Verified Carbon Units (VCUs) without any double counting for this monitoring period from 21-March-2021 to 30-Septemper-2022.

Besides, based on VVB's local expertise, China has a national emissions trading scheme only cover the high-emission industries, such as power generation sector that emitted at least 26,000 tons of CO_2e/year which has been verified in the public website (https://www.mee.gov.cn/xxgk2018/xxgk/xxgk02/202101/t20210105_816131.html), and it is confirmed that the project activity is not included the mandatory emission control scheme and there is no emission cap enforced for the project owner by checking the enforced company list in public information (https://www.mee.gov.cn/xxgk2018/xxgk/xxgk03/202012/t20201230_815546.html). Hence, it is confirmed that the emission reductions will not be double counted.

The project is seeking registration only under VCS program, with a 7-year renewable crediting period started from 21-March-2021 to 20-March-2028 (both days included) under VCS Program.

This monitoring period (21-March-2021 to 30-Septemper-2022) is the 1st monitoring period in the 7-year renewable crediting period under VCS program.

Via on-site inspection, checking the documents and interview with the project owner, CCSC verified that the project contributes the sustainable development goal through the following aspects

Table 4-1 Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7.2.1: Renewable energy share in the total final energy consumption	Implemented activities to increase	17,296.840 MWh of electricity from renewable sources have been supplied to the power grid during the first monitoring period (21-March-2021~30-Septemper-2022), replacing fossil fuel powered electricity.	From 21-March-2021 to 30-Septemper-2022, 17,296.840 MWh of electricity from renewable sources has been supplied to the power grid.

2)	8.5	The number of the job opportunities provided by the implementation of the project	Implemented activities to increase	Provided 19 employment positions for local people. No further changes this reporting period.	Provided 19 employment positions for local people in the reporting period.
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	The project achieved a GHG emission reduction of 47,385 tCO_2e during the first monitoring period(21-March-2021~30-Septemper-2022).	From 21-March-2021 to 30-Septemper-2022, the project has achieved a GHG emission reduction of 47,385 tCO_2e.

According to the information collected during on-site visit and the relevant evidence provided by project owner, CCSC can confirm that:

a) The monitoring period from 21-March-2021 to 30-Septemper-2022 is compliance with the crediting period and complies with the requirement of the VCS Standard Version 4.4 /5/;

b) No project design change occurred during this monitoring period and the project was implemented in accordance the project description in Joint PD-MR V02/3/;

c) The monitoring plan is implemented completely and monitoring system (i.e., process and schedule for obtaining, recording, compiling and analyzing the monitored data and parameters) is appropriate;

d) The monitoring system is in accordance with the monitoring plan set out in the project description and applied methodology AMS-III.D (ver.21.0) and AMS-I.D (ver.18.0);

e) The project has contributed to sustainable development during this monitoring period and over project lifetime;

f) The GHG emission reductions or removals generated by the project have not included in an emissions trading program or any other mechanism that includes GHG allowance trading;

g) The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period. None of the credits will be claimed in the same period;

h) The GHG emission reductions generated by the project during this monitoring period (21-March-2021 to 30-Septemper-2022) will be only claimed under VCS program as VCUs for the project to avoid double counting.

i) Data and parameter values used in the joint PD-MR are considered reasonable in the context of the project.

j) All estimates of baseline emissions can be replicated using the data and parameter values provided in the joint PD-MR.

h) The GHG ERRs have been quantified correctly in accordance with the project description and applied methodology.

4.1.1 Data and parameters fixed ex ante

The data and parameters fixed ex-ante are reported in the Joint PD-MR /3/, which have been checked against the monitoring plan and applied methodology /6//7/ by the assessment team.

Data / Parameter	GWP_{CH4}
Data unit	tCO_2e/tCH_4
Description	Global Warming Potential (GWP) of CH_4 applicable to the crediting period
Source of data	IPCC
Value applied	28
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-March-2021 to 20-March-2028 (renewable)) and thus is applicable to this monitoring period (21-March-2021 to 30-Septemper-2022). The value applied in the Joint PD-MR /3/ and ER calculation spreadsheet /4/ has been verified against IPCC and confirmed as consistent.

Data / Parameter	D_{CH4}
Data unit	kg/m^3
Description	CH_4 density
Source of data	AMS-III.D
Value applied	0.67 (at 20 ºC and 1 atm pressure)

Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-March-2021 to 20-March-2028 (renewable)) and thus is applicable to this monitoring period (21-March-2021 to 30-Septemper-2022). The value applied in the Joint PD-MR /3/ and ER calculation spreadsheet/4/ has been verified against AMS-III.D (Ver 21.0) and confirmed as consistent.

Data / Parameter	UF_b
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.D
Value applied	0.94
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-March-2021 to 20-March-2028 (renewable)) and thus is applicable to this monitoring period (21-March-2021 to 30-Septemper-2022). The value applied in the Joint PD-MR /3/ and ER calculation spreadsheet/4/ has been verified against AMS-III.D (Ver 21.0) and confirmed as consistent.

Data / Parameter	MCF_j
Data unit	-
Description	Annual methane conversion factor (MCF) for the baseline animal manure management system j
Source of data	2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	76%
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The value applied in the Joint PD-MR/3/ has been verified against the 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories and confirmed as consistent. According to para 18 and section 5 of Methodology AMS-III.D. the project, country-specific MCF values that reflect the specific management systems used in particular countries or regions are not available, thus, the IPCC default values provided in table

	10.17 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 are used and selected by temperature, ratio of potential evapotranspiration to precipitation and climate zone. According to official data published by local government on the website of government[3], the annual average temperature of baseline site where anaerobic manure treatment facility is located is around 13°C and ratio of potential evapotranspiration to precipitation is less than 1. The climate zone is warm temperature dry. Thus, the corresponding annual methane conversion factor (MCF) for the project is 76%.

Data / Parameter	$B_{0,LT}$
Data unit	m^3 CH_4/kg-VS
Description	Maximum methane producing potential of the volatile solid generated for animal type LT
Source of data	Table 10.16 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10
Value applied	Animal type: Dairy cattle; $B_{0,LT}$: 0.13
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The value has been verified against the 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4 Chapter 10 and confirmed as consistent. According to para 18 and section 5 of Methodology AMS-III.D the project, country specific B_o values are not available. Default values from table 10.16 of 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories volume 4 Chapter 10 is used. The species of the dairy cow in this project is Chinese Holstein which belongs to 'Asia' category. Value for $B_{0,\ LT,y}$ is 0.13 m^3CH_4/kg-VS.

[3] http://www.weixian.gov.cn/article/256/42972.html

Data / Parameter	$MS\%_{BL,i}$
Data unit	-
Description	Fraction of manure handled in baseline animal manure management system j
Source of data	FSR
Value applied	100%
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The value has been verified against FSR/25/ and confirmed as consistent with that of Joint PD-MR /3/.

Data / Parameter	$N_{da,y}$
Data unit	Day
Description	Number of days animal is alive in the farm in the year y
Source of data	Project proponents
Value applied	365
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The value applied has been verified through on-site interviewing with project owner and confirmed as correct.

Data / Parameter	$VS_{default}$
Data unit	kg/hd/day
Description	Default value for the volatile solid excretion rate per day on a dry-matter basis for a defined livestock population
Source of data	2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	Animal type / $VS_{default}$

	Dairy cattle | 3.474
Findings	According to para 18 and section 5 of Methodology AMS-III.D,, in China, there is no available country specific VS values. The parameters which the enhanced characterisation method (tier 2) needed are not available either. Thus, it is calculated via equation 10.22A in 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories. $VS_{default}$ =$VS_{rate(T)}$ * TAM / 1000 = 9.0 * 386 / 1000 TAM: Default Values for livestock weight for Animal Categories in Asia area is 386kg(mean) for dairy cattle $VS_{rate(T)}$ = Default Values for Volatile Solid Excretion Rate, 9.0 kg VS 1000 kg animal mass-1 day-1 for dairy cattle sourced from Volume 4, chapter 10, table 10.13A of 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories. The species of the dairy cattle in this project is Chinese Holstein. According to the staff in the pasture related to this project i.e. Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture, the average weight of the cow is around 700kg to 775kg which is far heavier than the default weight: 386kg according to IPCC default value. Heavier weight normally leads to a higher value of $VS_{LT,y}$ and a higher value of emission reduction as a result. Thus, using default value in IPCC is a conservative choice for this project. In addition, according to public available data, for Chinese Holstein species, the weight of bulls is usually from 800kg to 1000 kg; the weight of cows is usually from 450kg to 750 kg. The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The value has been verified against 2019 refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories

Data / Parameter	$EF_{grid,OM,y}$
Data unit	t CO_2/MWh
Description	Operating margin CO_2 emission factor in year y
Source of data	"2019 Baseline Emission Factors for Regional Power Grids in China" published by Ministry of Ecology and Environment of China
Value applied	0.9419

Purpose of Data	Calculation of baseline emissions and project emissions
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The value applied has been confirmed correct by checking "2019 Baseline Emission Factors for Regional Power Grids in China"/28/.

Data / Parameter	$EF_{grid,BM,y}$
Data unit	t CO_2/MWh
Description	Build margin CO_2 emission factor in year y
Source of data	"2019 Baseline Emission Factors for Regional Power Grids in China" published by Ministry of Ecology and Environment of China
Value applied	0.4819
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The value applied has been confirmed correct by checking "2019 Baseline Emission Factors for Regional Power Grids in China"/28/.

Data / Parameter	$EF_{grid,CM,y}$ ($EF_{grid,y}$, $EF_{EL,j,y}$)
Data unit	t CO_2/MWh
Description	Combined margin emission factor for the grid in year y
Source of data	Calculated based on "Tool to calculate the emission factor for an electricity system" (version 07.0).
Value applied	0.7119
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The calculation of the applied value has been verified against Tool to calculate the emission factor for an electricity system" (version 07.0)/8/.

Data / Parameter	NCV_{CH4}
Data unit	MJ/Nm^3
Description	NCV of methane
Source of data	AMS-III.D
Value applied	35.9
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The applied value has been confirmed against AMS-III.D (ver 21.0).

Data / Parameter	$TDL_{j,y}$
Data unit	-
Description	Average technical transmission and distribution losses for providing electricity to source j in year y
Source of data	Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)
Value applied	20%
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The applied value has been verified against Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation (version 03.0)/10/ and confirmed as consistent.

Data / Parameter	EE_y
Data unit	%
Description	Energy Conversion Efficiency of the project equipment
Source of data	AMS-III.D
Value applied	40%

Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The value has been verified against AMS-III.D and confirmed as consistent.

Data / Parameter	$SPEC_{flare}$
Data unit	Temperature - °C Flow rate or heat flux - kg/h or m^3 /h Maintenance schedule - number of days
Description	Manufacturer's flare operating specifications for temperature, flow rate and maintenance schedule
Source of data	Flare manufacturer
Value applied	Temperature – 500~1270°C Flow rate or heat flux - 1100 m^3 /h Maintenance schedule - every 12-24 months (around 365days – 730 days) based on operating conditions
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The applied value has been verified against flare manufacturer and confirmed as consistent.

Data / Parameter	$\eta_{flare,m}$
Data unit	%
Description	Flare efficiency in minute m
Source of data	Default value of methodological tool 06" Project emissions from flaring (Version 04.0)"
Value applied	When the flame is detected in the minute m and the temperature of the flare and the flow rate of the residual gas to the flare ($F_{RG,m}$) is within the manufacturer's operating specification for the flare ($SPEC_{flare}$) in the minute m: 90% Otherwise: 0%
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-

	Mar-2021 to 30-Sep-2022). The applied value has been verified against default value of methodological tool 06" Project emissions from flaring (Version 04.0)" .

Data / Parameter	$f_{CH4,default}$
Data unit	m^3 CH_4 / m^3 biogas corrected to reference conditions
Description	Default value for the fraction of methane in the biogas
Source of data	The default value was derived based on reported values from registered projects and research papers (Davidsson, 2007)
Value applied	0.6
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The applied value has been verified against the default value was derived based on reported values from registered projects and research papers (Davidsson, 2007).

Data / Parameter	R_u
Data unit	Pa.m^3/kmol.K
Description	Universal ideal gases constant
Source of data	"Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)
Value applied	8,314
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The applied value has been verified against "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)

Data / Parameter	MM_i
Data unit	kg/kmol
Description	Molecular mass of greenhouse gas CH_4
Source of data	"Tool to determine the mass flow of a greenhouse gas in a gaseous

	stream" (Version 03.0)
Value applied	16.04
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The applied value has been verified against the default value was derived based on reported values from registered projects and research papers (Davidsson, 2007).

Data / Parameter	P_n
Data unit	Pa
Description	Total pressure at normal conditions
Source of data	"Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)
Value applied	101,325 Pa
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The applied value has been verified against "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)

Data / Parameter	T_n
Data unit	K
Description	Temperature at normal conditions
Source of data	"Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)
Value applied	273.15
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The applied value has been verified against "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)

Data / Parameter	$\rho_{i,n}$ & $\rho_{RG,n}$
Data unit	Kg gas i/m³ gas I, i=RG=CH_4
Description	Density of greenhouse gas CH_4 in the gaseous stream at normal conditions (0℃ and 1 atm)
Source of data	Calculated in the section 3.4.6 as per the latest version of "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)
Value applied	0.71566
Findings	The Parameter has been determined ex-ante in the monitoring plan for this crediting period (21-Mar-2021 to 20-Mar-2028 (renewable)) and thus is applicable to this monitoring period (21-Mar-2021 to 30-Sep-2022). The applied value has been verified against calculated in the section 3.4.6 as per the latest version of "Tool to determine the mass flow of a greenhouse gas in a gaseous stream" (Version 03.0)

In conclusion, according to VCS criteria, and based on the assessment team's local and sectorial knowledge, the assessment team confirms that:

The data and parameters fixed ex-ante have been correctly listed. The parameters fixed ex-ante have been verified by checking the information flow and in compliance with the monitoring plan of the Joint PD-MR.

4.1.2 Data and parameters monitored

According to VCS criteria, the assessment team has performed the following activities to determine whether the monitoring of parameters related to the GHG emission reductions has been implemented in accordance with the monitoring plan.

Through the inspection of the monitoring system, interview with the operation staff, document review including relevant records, procedures and technical specifications, the assessment team has assessed the implementation of the monitoring plan followed by the PP;

The parameters stated in the monitoring plan have been checked by means above;

The assessment team has checked the installation of the electricity meter against PPA, diagram of power connection system and calibration certificate of electricity meter by qualified third party/30/;

The Monitoring data /23/ and Electricity transaction note/24/ were checked by the team to confirm the monitoring results;

Through interviewing with the top management and operation staff, the assessment team has assessed the quality assurance and quality control procedures applied by the PP.

Data / Parameter	$MS\%_{i,y}$
Data unit	-
Description	Fraction of manure handled in system i in year y
Value applied:	100%
Findings	By on-site inspection and checking the FSR, the assessment team confirmed the applied value as correct

Data / Parameter	$N_{p,y}$
Data unit	number
Description	Number of animals produced annually of type LT for the year y
Value applied:	<table><tr><td>Period</td><td>Value(head)</td></tr><tr><td>21-Mar-2021~31-Dec-2021</td><td>12,059</td></tr><tr><td>01-Jan-2022~30-Sep-2022</td><td>12,093</td></tr></table>
Findings	Via on-site inspection, the assessment team confirmed that this parameter is recorded monthly according to daily report of cattle dynamics from Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture and aggregated annually, which is consistent with the monitoring plan described in the Joint PD-MR/1//3/. The assessment team has checked the report of cattle dynamics/33/ and the description from Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture, and confirmed that the reported monitoring data is consistent with the documented evidences.

Data / Parameter	nd_y
Data unit	number
Description	The number of days the treatment plant was operational in year y
Value applied:	

	<table><tr><td>Period</td><td>Value(day)</td></tr><tr><td>21-Mar-2021~31-Dec-2021</td><td>285</td></tr><tr><td>01-Jan-2022~30-Sep-2022</td><td>273</td></tr></table>
Findings	Via checking the operating records during site visit, it is confirmed that the data was monitored as described in the monitoring plan in the Joint PD-MR/1//3/. The assessment team has checked the data included in the Joint PD-MR version 02.0 /3/and ER calculation spreadsheet /4/against the monthly operating records/23/ and can confirm that the reported monitoring data is consistent with the documented evidences.

Data / Parameter	EG_y
Data unit	MWh
Description	Total electricity generated from the recovered biogas in year y
Value applied:	-
Findings	The parameter is not monitored by the project owner, it has been replaced by the parameter $EG_{PJ,y}$. EG_y is the total electricity generated from the recovered biogas in year y. $EG_{PJ,y}$ is the quantity of electricity generation supplied by the project to the grid in year y. Using $EG_{PJ,y}$ to replace EG_y is more conservative. In addition, no matter it is EG_y or $EG_{PJ,y}$, it does not influence the amount of the final emission reduction since after the minimization *according to the* methodology of $\left(\mathrm{BE}_{\mathrm{y,ex\ post}} - \mathrm{PE}_{\mathrm{y,ex\ post}}\right)\mathrm{and}\left(\mathrm{MD}_{\mathrm{y}} - \mathrm{PE}_{\mathrm{power,y,ex\ post}}\right)$, the value of $\left(\mathrm{BE}_{\mathrm{y,ex\ post}} - \mathrm{PE}_{\mathrm{y,ex\ post}}\right)$ is taken. The assessment team has checked the data included in the Joint PD-MR version 03/38/and ER calculation spreadsheet /4/against the monthly operating records/23/ and confirmed that the reported monitoring data is consistent with the documented evidences.

Data / Parameter	$EC_{PJ,j,y}$
Data unit	MWh
Description	Quantity of electricity consumed by the project in year y

Value applied:	<table><tr><th>Period</th><th>$EC_{PJ,j,y}$ (MWh)</th></tr><tr><td>21-Mar-2021~31-Dec-2021</td><td>0.000</td></tr><tr><td>01-Jan-2022~30-Sep-2022</td><td>0.000</td></tr><tr><td>Total</td><td>0.000</td></tr></table>
Findings	During this monitoring period, the data is zero and it is confirmed by checking the reading records, electricity transaction note/24/ and interview the staff of the project. The metering instruments has been calibrated annually in accordance with the latest version “Technical administrative code of electric energy metering”. The information of electricity meter: The electricity meter is at the project site between the project and NCPG. This parameter is monitored continuously and recorded monthly. The cut-off time of the electricity meter E is at 24:00 of the last day of each month. <table><tr><th>Type</th><th>Serial number</th><th>Accuracy class</th><th>Date of calibration</th><th>Validity</th></tr><tr><td>DSZ178</td><td>4300036005</td><td>0.2S</td><td>22-Dec-2020;20-Dec-2021</td><td>21-Dec-2021;19-Dec-2022</td></tr></table>The assessment team has checked the data included in the Joint PD-MR version 02.0/3/and ER calculation spreadsheet /4/against the monthly operating records/23/ and confirmed that the reported monitoring data is consistent with the documented evidences.

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh
Description	Quantity of electricity generation supplied by the project to the grid in year y
Value applied:	<table><tr><th>Period</th><th>$EG_{PJ,y}$(MWh)</th></tr><tr><td>21-Mar-2021~31-Dec-2021</td><td>7,739.360</td></tr><tr><td>01-Jan-2022~30-Sep-2022</td><td>9,557.480</td></tr></table>

	Total | 17,296.840
Findings	The data is cross checked with Electricity Transaction Note/24 /. The metering instruments has been calibrated annually during this monitoring period, which is in accordance with the monitoring plan of the Joint PD-MR/3/. The information of electricity meter: The electricity meter is at the project site between the project and NCPG. This parameter is monitored continuously and recorded monthly. The cut-off time of the electricity meter E is at 24:00 of the last day of each month. Type | Serial number | Accuracy class | Date of calibration | Validity DSZ178 | 4300036005 | 0.2S | 22-Dec-2020;20-Dec-2021 | 21-Dec-2021;19-Dec-2022 The assessment team has checked the data included in the Joint PD-MR version 02.0 /3/and ER calculation spreadsheet/4/against the monthly operating records/20/ and can confirm that the reported monitoring data is consistent with the documented evidences.

Type	Serial number	Accuracy class	Date of calibration	Validity
DSZ178	4300036005	0.2S	22-Dec-2020;20-Dec-2021	21-Dec-2021;19-Dec-2022

Data / Parameter	$F_{CH4,RG,m}$
Data unit	t CH_4
Description	Mass flow of methane in the residual gas in the minute m
Value applied:	Calculated
Findings	The assessment team has checked the data included in the Joint PD-MR version 03.1/39/and confirmed that there is no malfunction during this verification period

Data / Parameter	$V_{RG,t,db}$
Data unit	m^3 dry gas/h
Description	Volumetric flow of the residue gas in time interval t on a dry basis at normal conditions
Value applied	The data will be monitored ex post in the monitoring period. In the

	estimated net GHG emission reduction phase, the amount of biogas is estimated and calculated as per section 4, please refer to ER calculation sheet for details.
Findings	<table><tr><td>Type</td><td>Serial number</td><td>Date of calibration</td><td>Validity</td></tr><tr><td>HGF-3000F</td><td>200051040051</td><td>06-Apr-2020;05-Apr-2022</td><td>05-Apr-2022;04-Apr-2024</td></tr></table> The data will be monitored ex post in the monitoring period. In the estimated net GHG emission reduction phase, the amount of biogas is estimated and calculated as per section 4, please refer to ER calculation sheet for details.Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory for all projects applying large scale methodology(ies). Calibration will be carried out once every two years according to manufacturer's specifications. Periodic calibration against a primary device provided by an independent accredited laboratory is mandatory for all projects applying large scale methodology(ies). Calibration will be carried out once every two years according to manufacturer's specifications. The assessment team has checked the data included in the Joint PD-MR version 03.1/39/ and the data about the specifications, models and verification certificates ot the flow meter/40/ and confirmed that the verification certificate of the flowmeter is within the validity period， and there is no malfunction during this verification period.

Data / Parameter	Operation hours (to calculate $F_{CH4,RG,m}$)
Data unit	Hours
Description	The total operation hours of the flare (to calculate $F_{CH4,RG,m}$)
Value applied	The data will be monitored ex post in the monitoring period.
Findings	The assessment team has checked the data included in the Joint PD-MR version 03.1/39/and confirmed that there is no malfunction during this verification period

In conclusion, CCSC confirms that:

- The data and parameters fixed ex-ante have been correctly listed. Parameters fixed ex-ante for required parameters have been verified by checking the information flow and in compliance with the monitoring plan of the Joint PD-MR./3/

- The monitoring has been carried out in accordance with the monitoring plan contained in Joint PD-MR /3/.
- All parameters required by the monitoring plan have been sufficiently monitored and correctly listed. The monitored data for required parameters have been verified by checking the whole information flow.
- The calibration of the electricity meter has been conducted at the frequency as specified by the methodologies and the monitoring plan contained in the Joint PD-MR/3/.

4.1.3 Assessment of data and calculation of emission reductions

Baseline emissions

Baseline emissions are calculated using equation below:

$$BE_y = BE_{CH4,y} + BE_{Elec,y}$$

1.Baseline emissions from the manure treatment processes in year y ($BE_{CH4,y}$)

According to AMS-III.D, the emission reductions achieved in any year due to utilization of manure are the lowest value of the following:

$$\mathrm{ER_{y,ex\ post}} = \min\left[\left(\mathrm{BE_{y,ex\ post}} - \mathrm{PE_{y,ex\ post}}\right), \left(\mathrm{MD_y} - \mathrm{PE_{power,y,ex\ post}}\right)\right]$$

$BE_{y,ex\ post}$ is calculated according to the equation below:

$$BE_{y,ex\ post} = GWP_{CH4} \times D_{CH4} \times UF_b \times \sum\nolimits_{j,LT} MCFj \times B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y}$$

The value of $BE_{y,\ ex\ post}$ monitored during this monitoring period is shown in table 4-2.

Table 4-2 The value of $BE_{y,\ ex\ post}$ during this monitoring peiod

Period	21-Mar-2021~31-Dec-2021	01-Jan-2022~30-Sep-2022	Unit
GWP_{CH4}	28	28	tCO_2e/tCH_4
D_{CH4}	0.00067	0.00067	t/m^3
UF_b	0.94	0.94	-
MCF_j	76%	76%	
$B_{0,LT}$	0.13	0.13	
$N_{LT,y}$	12,059	12,093	hd
$VS_{LT,y}$	990.09	948.40	kg/hd/day
$MS\%_{i,y}$	100%	100%	
$BE_{y,ex\ post}$	20,801	19,982	tCO_2e

MD_y is calculated using the equation below:

$$MD_y = \frac{EG_y \times 3600}{NCV_{CH4} \times EE_y} \times D_{CH4} \times GWP_{CH4}$$

For this project, the EG_y has been replaced by $EG_{PJ,y}$ and the monitored value of $EG_{PJ,y}$ is shown in the table 4-3.

Table 4-3 The monitored value of $EG_{PJ,y}$

Period		$EG_{PJ,y}$(MWh)		
		Meter Reading Record	Electricity Transaction Note Record	Conservative Data
21-Mar-2021	30-Apr-2021	701.800	700.320	700.320
01-May-2021	31-May-2021	630.400	629.800	629.800
01-Jun-2021	30-Jun-2021	721.800	721.040	721.040
01-Jul-2021	31-Jul-2021	722.200	721.200	721.200
01-Aug-2021	31-Aug-2021	681,000	679.800	679.800
01-Sep-2021	30-Sep-2021	969,300	968.240	968.240
01-Oct-2021	31-Oct-2021	1,184,400	1,183.360	1,183.360
01-Nov-2021	30-Nov-2021	1,111,500	1,110.440	1,110.440
01-Dec-2021	31-Dec-2021	1,026,000	1,025.160	1,025.160
Subtotal				7,739.360
01-Jan-2022	31-Jan-2022	1,058.200	1,057.000	1,057.000
01-Feb-2022	28-Feb-2022	1,052.800	1,051.600	1,051.600
01-Mar-2022	31-Mar-2022	1,206.600	1,204.920	1,204.920
01-Apr-2022	30-Apr-2022	1,097.000	1,094.960	1,094.960
01-May-2022	31-May-2022	1,070.000	1,068.680	1,068.680
01-Jun-2022	30-Jun-2022	914.700	912.440	912.440
01-Jul-2022	31-Jul-2022	985.400	984.200	984.200
01-Aug-2022	31-Aug-2022	969.600	967.760	967.760
01-Sep-2022	30-Sep-2022	1,218.200	1,215.920	1,215.920
Subtotal				9,557.480
Total				17,296.840

The value of MD_y monitored during this monitoring period is shown in the table 4-4

Table 4-4 The value of MD_y during this monitoring period

Period	$EG_{PJ,y}$(MWh)	GWP_{CH4}(tCO_2e/tCH_4)	D_{CH4}(kg/m^3)	NCV_{CH4} (MJ/Nm^3)	EE_y	MDy(tCO_2e)
	A	B	C	D	E	F=A*3600/(D*E)*C*B*0.001
21-Mar-2021~31-Dec-2021	7,739.360	28	0.67	35.9	0.4	36,398

01-Jan-2022~30-Sep-2022	9,557.480	28	0.67	35.9	0.4	44,949
Total	17,296.840	/	/	/	/	81,347

2. Baseline emissions associated with electricity generation in year y ($BE_{elec,y}$)

As per AMS-I. D, baseline emissions associated with electricity generation in year y is calculated according to equation below:

$$BE_{elec,y} = EG_{PJ,y} \times EF_{grid,y}$$

The value of $BE_{elec,y}$ monitored during this monitoring period is shown in the table 4-5

Table 4-5 The value of $BE_{elec,y}$ during this monitoring period

Period	$EG_{PJ,y}$(MWh)	$EF_{grid,CM,y}$ (t CO_2/MWh)	$BE_{Elec,y}$
	A	B	C=A*B
21-Mar-2021~31-Dec-2021	7,739.360	0.7119	5,509
01-Jan-2022~30-Sep-2022	9,557.480	0.7119	6,803
Total	17,296.840	/	12,312

Project emissions

For this project, the $PE_{y,\ ex\ post}$ is calculated using the equation below:

$$PE_y = PE_{PL,y} + PE_{EC,y}$$

1. Emissions due to physical leakage of biogas

$$PE_{PL,y} = 0.10 \times GWP_{CH4} \times D_{CH4} \times \sum_{i,LT} B_{0,LT} \times N_{LT,y} \times VS_{LT,y} \times MS\%_{i,y}$$

The value of $PE_{PL,y}$ monitored during this monitoring period is shown in the table 4-6

Table 4-6 The value of $PE_{PL,y}$ monitored during this monitoring period

Period	21-Mar-2021~31-Dec-2021	01-Jan-2022~30-Sep-2022	Unit
/	0.10	0.10	/
GWP_{CH4}	28	28	tCO_2e/tCH_4
D_{CH4}	0.00067	0.00067	t/m^3
$B_{0,LT}$	0.13	0.13	
$N_{LT,y}$	12,059	12,093	hd

$VS_{LT,y}$	990.09	948.40	kg/hd/day
$MS\%_{i,y}$	100%	100%	
$PE_{PL,y}$	2,912	2,798	tCO_2e

2. Emissions from electricity consumption

$$PE_{EC,y} = \sum_j EC_{PJ,j,y} \times EF_{EL,j,y} \times (1 + TDL_{j,y})$$

For this project, the electricity generated by the project are used for self-consumption first and then supplied to the grid. The extra electricity consumption will only happen when all the generators stop running. During this monitoring period, according to the record, no such situation happened, the extra electricity consumed by the project, $EC_{PJ,j,y}$ =0.

Thus, $PE_{power,\ y,\ ex\ post}$ = $PE_{EC,y}$ =0

3. Emissions from flaring or combustion of the biogas stream in the year y

There is no malfunction and the flare system has not been used during this monitoring period. $PE_{flare,y}$ = 0.

Leakage

Leakage for this project is 0.

Net GHG Emission Reductions and Removals

$ER_{y,\ ex\ post}$ is calculated during this monitoring period using the equation and the value is shown in table 4-7 below:

$$ER_{y,ex\ post} = \min[(BE_{y,ex\ post} - PE_{y,ex\ post}),(MD_y - PE_{power,y,ex\ post})]$$

Table 4-7 The value of $ER_{y,\ ex\ post}$ during this monitoring period

Period	$BE_{y,ex\ post}$	$PE_{y,\ ex\ post}$	$BE_{y,ex\ post}$ - $PE_{y,\ ex\ post}$	MD_y	$PE_{power,y,ex\ post}$	MD_y - $PE_{power,y,ex\ post}$	$ER_{y,ex\ post}$
	A	B	C=A-B	D	E	F=D-E	G=min(C,F)
21-Mar-2021~31-Dec-2021	20,801	2,912	17,889	36,398	0	36,398	17,889
01-Jan-2022~30-Sep-2022	19,982	2,798	17,184	44,949	0	44,949	17,184
Total							35,073

Table 4-8 The total baseline emission reductions for the project are calculated

Period	$BE_{y,ex\ post}$	$BE_{Elec,y}$	BE
	A	B	C=A+B
21-Mar-2021~31-Dec-2021	20,801	5,509	26,310

01-Jan-2022~30-Sep-2022	19,982	6,803	26,785
Total	40,783	12,312	53,095

Thus , the calculation of the net GHG Emission Reductions and Removals in this monitoring period is summarized in the following table:

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
21-Mar-2021~31-Dec-2021	26,310	2,912	0	23,398
01-Jan-2022~30-Sep-2022	26,785	2,798	0	23,987
Total	53,095	5,710	0	47,385

The emission reductions during the monitoring period are calculated as: $\mathrm{ER_y} = \mathrm{BE_y} - \mathrm{PE_y} - \mathrm{LE_y}$ =53,095−5,710−0=47,385 tCO_2e

This monitoring period is from 21-Mar-2021 to 30-Sep-2022. Based on the annual estimated emission reductions, the amount of emission reductions for this monitoring period would be 87,633 tCO_2e. The actual emission reductions in this monitoring period are 47,385 tCO_2e, which is 45.93% lower than the estimation. One reason is that the actual installed capacity of the project during this monitoring period is 2.33MW (2 sets of 1.165MW CMM generators have been operated to generate electricity), while the total intended installed capacity of the project is 3.83MW, the estimated emission reductions for this monitoring period was re-calculated based on the actual operation capacity. The re-calculated estimated emission reductions for this monitoring period are 53,312 tCO_2e (=87,633 tCO_2e × 2.33MW / 3.83MW). The actual emission reductions achieved in this monitoring period is 11.12% lower than the re-calculated estimated emission reductions. Considering the insufficient manure wastes supply which leads to insufficient biogas, the difference is reasonable.

Corresponding to the VCS criteria, CCSC assessment team confirms that:

- A complete set of data for the monitoring period is available.
- Information provided in the Joint PD-MR /3/ has been cross-checked with other sources;

- Calculations of baseline emissions, and project activity emissions and leakage have been appropriately carried out in accordance with the formulae and methods described in the monitoring plan and the applied methodologies.
- There are no assumptions in emission reductions calculation.
- Appropriate parameters fixed ex ante of the project has been correctly applied.
- A comparison of actual GHG emission reductions or net anthropogenic GHG removal of the project activity achieved during this monitoring period with the estimates in the Joint PD-MR has been provided, and the results are correct.

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

As a result of verification of the ER calculation process, the assessment team confirmed that all the parameters required for the determination of the emission reductions have been included in the Joint PD-MR/3/ and ER calculation spreadsheet /4/ of this monitoring period and are consistent with the applied methodology AMS-III.D Methane recovery in animal manure management systems (version 21.0)/6/&AMS-I.D Grid connected renewable electricity generation (version 18.0)/7/., and the monitoring plan contained in the Joint PD-MR. The parameters are complete in this monitoring period.

After verifying the reported values with the raw data sources, it's confirmed that the values of the parameters from the raw data sources are consistent with those quoted in the ER Calculation Spreadsheet of this monitoring period and Joint PD-MR. The verification process for the same has been clearly described above in section 4.2 of the report.

All necessary documentations are collected, referenced and aggregated, which is easily accessible in hard-copy or electronic format. Measurements are performed by calibrated equipment, and the key data can also be cross-checked via other sources, such as records, receipts and inventory data. No assumptions are used that have any material influence on reported emission reductions.

CCSC concludes that during this monitoring period, the evidences for determination of emission reductions are sufficient and reasonable, and the calculation of emission reductions is reliable.

5 VALIDATION AND VERIFICATION OPINION

China Classification Society Certification Co. Ltd. (CCSC) has performed the validation of Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture Wastes Recycling Project and the verification of the project covering the monitoring period from 21-Mar-2021 to 30-Sep-2022 within the renewable crediting period from 21-Mar-2021 to 20-Mar-2028 (both days included), on the basis of requirements of Verified Carbon Standard (VCS) Version 4.4.

The review of the project design documentation and the subsequent follow-up interviews have provided CCSC with sufficient evidence to determine the fulfilment of stated criteria.

In the course of the validation and verification, 0 Clarification Request (CL), 2 Corrective Action Requests (CARs) and 0 Forward Action Request (FAR) were raised for the project activity (Joint PD-MR Version 01, dated 29-December-2022/1/) in relation to all relevant VCS requirements. Until issuance of this version of report, the raised CARs were successfully closed.

The review of the project design documentation and additional documents related to baseline and monitoring methodology and subsequent background investigation have provided the CCSC with sufficient evidence to validate the fulfilment of the stated criteria. In detail the conclusions can be summarized as follows:

- The baseline scenario is correctly defined as per the applied methodology and relate tools;
- The project is additional;
- All data and information used for ex-ante calculation of emission reductions is of projected and/or hypothetical nature;
- A reasonable level of assurance has been applied;
- The monitoring plan in the validated Joint PD-MR/3/ is as per the applied methodology AMS-III.D(version 21.0) and AMS-I.D(version 18.0);
- The project has been implemented and operated as per the Joint PD-MR;
- The project complies with the validation criteria for projects set out in VCS standard Version 4.4;
- The calculation of the ex-ante emission reductions is carried out in a transparent and conservative manner, so that the calculated emission reductions of 400,540 tCO_2e are most likely to be achieved within the 7-year renewable crediting period started from 21-Mar-2021.

It is CCSC's responsibility to provide an independent validation statement on the Project's compliance with the VCS Program requirements as set out in the VCS Program Rules. The conclusions of this report show that the project, as it was described in the project description contained in the Joint PD-MR, is in line with all criteria applicable for the validation against the VCS standard version 4.4 without any qualifications or limitations.

The verification is based on the baseline and monitoring methodology AMS-III.D(version 21.0) and AMS-I.D(version 18.0), and the Joint PD-MR (version 02.0 dated 06-March-2023)/3/. The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the monitoring plan and the reporting of GHG emission reductions on the basis set out within the Joint PD-MR. Our verification approach was based on the requirements as defined under the applicable VCS Version 4 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- The monitoring plan in Joint PD-MR is as per the applied methodology;
- The monitoring complies with the monitoring plan in the Joint PD-MR;
- The Joint PD-MR and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS and CDM requirements;
- The installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- The monitoring system is in place and generates GHG emission reductions data;
- The GHG emission reductions are calculated without material misstatements.

It is CCSC's responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CCSC planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

It is CCSC's responsibility to towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CCSC and the engagement conditions detailed in this report. CCSC cannot be held liable by any party for decisions made or not made based on this report.

In conclusion, it is CCSC's opinion that the project "Leyuan ANIMAL Husbandry Weixian Co., Ltd. NO. 4 Pasture Wastes Recycling Project", in the Joint PD-MR (version 02.0 dated 06-March-2023) meets all relevant requirements for VCS activities and all relevant host Party criteria and correctly applies methodologies "AMS-III.D Methane recovery in animal manure management systems (version 21.0)" and "AMS-I. D Grid connected renewable electricity generation (version 18.0)". And, CCSC confirms the following statement:

Verification period: From 21-Mar-2021 to 30-Sep-2022

Verified GHG emission reductions and removals in the above verification period:

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
21-Mar-2021~31-Dec-2021	26,310	2,912	0	23,398
01-Jan-2022~30-Sep-2022	26,785	2,798	0	23,987
Total	53,095	5,710	0	47,385

APPENDIX A:ABBREVIATIONS

CAR	Corrective Action Request
CCSC	China Classification Society Certification Co. Ltd.
NCPG	North China Power Grid
CDM	Clean Development Mechanism
CER	Certified Emission Reduction(s)
CL	Clarification request
CO_2	Carbon Dioxide
CO_2e	Carbon Dioxide Equivalent
DOE	Designated Operational Entity
DNA	Designated National Authority
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETN	Electricity Transaction Note
FAR	Forward Action Request
GHG	Greenhouse Gas(es)
Joint PD-MR	Joint Project Description & Monitoring Report
PD	Project Description
PDD	Project Design Document
PP	Project Proponent
UNFCCC	United Nations Framework Convention on Climate Change
VVB	Validation/Verification Body
VVS	Validation and Verification Standard for project activities
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit

APPENDIX B: REFERENCE

Ref no.	Reference Document
/1/	Climate Bridge (Shanghai) Ltd. Joint project description and monitoring report, Version 01 dated 29-Dec-2022
/2/	Climate Bridge (Shanghai) Ltd. ER calculation spreadsheet, Version 01 dated 29-Dec-2022
/3/	Climate Bridge (Shanghai) Ltd. Joint project description and monitoring report, Version 02.0 dated 06-March-2023
/4/	Climate Bridge (Shanghai) Ltd. ER calculation spreadsheet Version 02.0 dated 06-March-2023
/5/	VERRA, VCS Standard, version 4.4
/6/	CDM Executive Board, AMS-III.D(Version 21.0)
/7/	CDM Executive Board, AMS-I.D(Version 18.0)
/8/	CDM Executive Board, Tool to calculate the emission factor for an electricity system, (version 07.0)
/9/	CDM Executive Board, Project and leakage emissions from anaerobic digesters (version 02.0)
/10/	CDM Executive Board, Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0
/11/	CDM Executive Board, Positive lists of technologies, version 04.0
/12/	VERRA, VCS Program Guide Version 4.3
/13/	VERRA, Registration & Issuance Process Version 4.3
/14/	VERRA, Methodology-Requirements Version 4.3

/15/	Hebei Zhonghai Huaneng Energy Co., Ltd Business License
/16/	Xingtai Administrative Examination and Approval Bureau, Project approval dated 7-May-2020
/17/	Weixian Branch of Xingtai Ecological and Environmental Bureau ,EIA approval dated 4-Dec-2019
/18/	Nameplate of the equipment
/19/	Contract and annex for construction of Anaerobic System, signed between the PP and Beijing Yingherui Environmental Technology Co., Ltd
/20/	Sales contract of container-type biogas cogeneration unit, signed between the PP and Shanghai Qiyao Power Technology Co., Ltd
/21/	Technical Agreement for container-type biogas cogeneration unit Procurement
/22/	Power Purchase Agreement, signed between the PP and local Power Grid Company
/23/	Monthly records during this monitoring period
/24/	Electricity transaction note issued by the grid company during this monitoring period
/25/	Feasibility study report (FSR) dated July-2019
/26/	Environmental Impact Assessment Report dated Oct-2019
/27/	Power Grid Company, Grid-connected dispatching protocol
/28/	Chinese DNA, 2019 Baseline Emission Factors for Regional Power Grids in China dated 29-Dec-2020
/29/	Questionnaire survey, March 2021, June 2022
/30/	Calibration Certificate of electricity meter during this monitoring period
/31/	Qualification certificate of calibration entities: Hebei Institute of Metrology Supervision and Testing, valid from 19-Aug-2022 to 18-Aug-2027

/32/	Commencement and completion acceptance report
/33/	The record sheet and the description of cattle dynamic
/34/	Social security certificate of employees
/35/	The photo of electricity meter
/36/	VERRA, Vcs-Program-Definitions-v4.3
/37/	VCS-Joint-Project-Description-Monitoring-Report-Template-v4.2
/38/	Climate Bridge (Shanghai) Ltd. Joint project description and monitoring report, Version 03 dated 30-June-2023
/39/	Climate Bridge (Shanghai) Ltd. Joint project description and monitoring report, Version 03.1 dated 06-Nov-2023
/40/	Specifications, models and verification certificates ot the flow meter

APPENDIX C: RESOLUTION OF CLARIFICATION REQUESTS AND CORRECTIVE ACTION REQUESTS AND FORWARD ACTION REQUESTS

Table 1. CL from this validation and verification

CL ID	N/A	Section no.	N/A	Date: N/A
Description of CL				
N/A				
Project participant response				**Date:** N/A
N/A				
Documentation provided by project participant				
N/A				
DOE assessment				**Date:** N/A
N/A				

Table 2. CAR from this validation and verification

CAR ID	CAR-1	Section no.	1.11	Date: 04-March-2023
Description of CAR				
The rated power of the generator in the PD-MR (Version 01) is inconsistent with rated power indicated on the generator nameplate				
Project participant response				**Date:** 06-March-2023
The inconsistence of the generator between the description in Joint PD-MR/1/ and the nameplate is a slip of the pen. By checking the nameplate and site visiting, it has been confirmed that the rated power of the generator is 1.734MW, the same as the nameplate. The description in PD-MR has been corrected.				

Documentation provided by project participant				
Joint PD-MR (Version 02)				
DOE assessment				**Date:** 06-March-2023
The assessment team has checked the updated Joint PD-MR (Version 02). CCSC confirms the rated power of generator has been corrected. Therefore, CAR-1 is closed successfully.				
CAR ID	CAR-2	**Section no.**	7.2	**Date:** 04-March-2023
Description of CAR				
The meter reading record in the monitored value of EG_{pjy} sheet in the Joint PD-MR (Version 01.1) and in the ER calculation spreadsheet is inconsistent with the Monthly records during this monitoring period				
Project participant response				**Date:** 06-March-2023
After confirmation with on-site technical staff, the project proponent has corrected the description in Joint PD-MR according to the monthly records during this monitoring period. The calculation involved $EG_{PJ,y}$ has been checked and corrected accordingly.				
Documentation provided by project participant				
Joint PD-MR (Version 02.0)				
DOE assessment				**Date:** 06-March-2023
The assessment team has checked the Joint PD-MR (Version 02). CCSC confirms the information of the meter reading record in the monitored value of $EG_{PJ,y}$ sheet has been corrected. Therefore, CAR-2 is closed successfully.				

Table 3. FAR from this validation and verification

FAR ID	N/A	**Section no.**	N/A	**Date:** N/A
Description of FAR				
N/A				
Project participant response				**Date:** N/A
N/A				

Documentation provided by project participant	
N/A	
DOE assessment	**Date:** N/A
N/A	